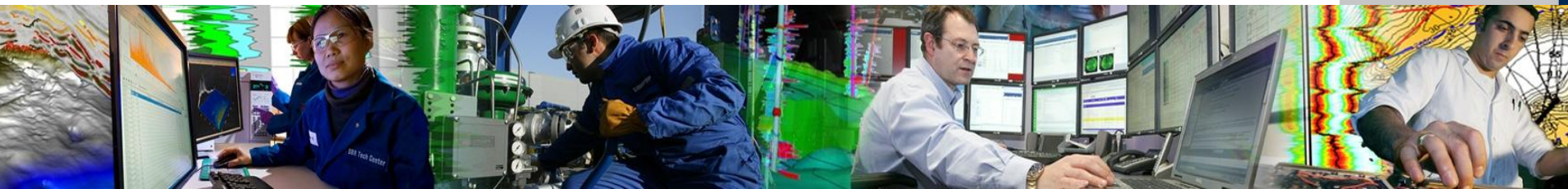


Mapping the Subsurface - with Petrel



Seismic Interpretation Fundamentals for mapping

Integration of Geophysical Data in Subsurface Mapping

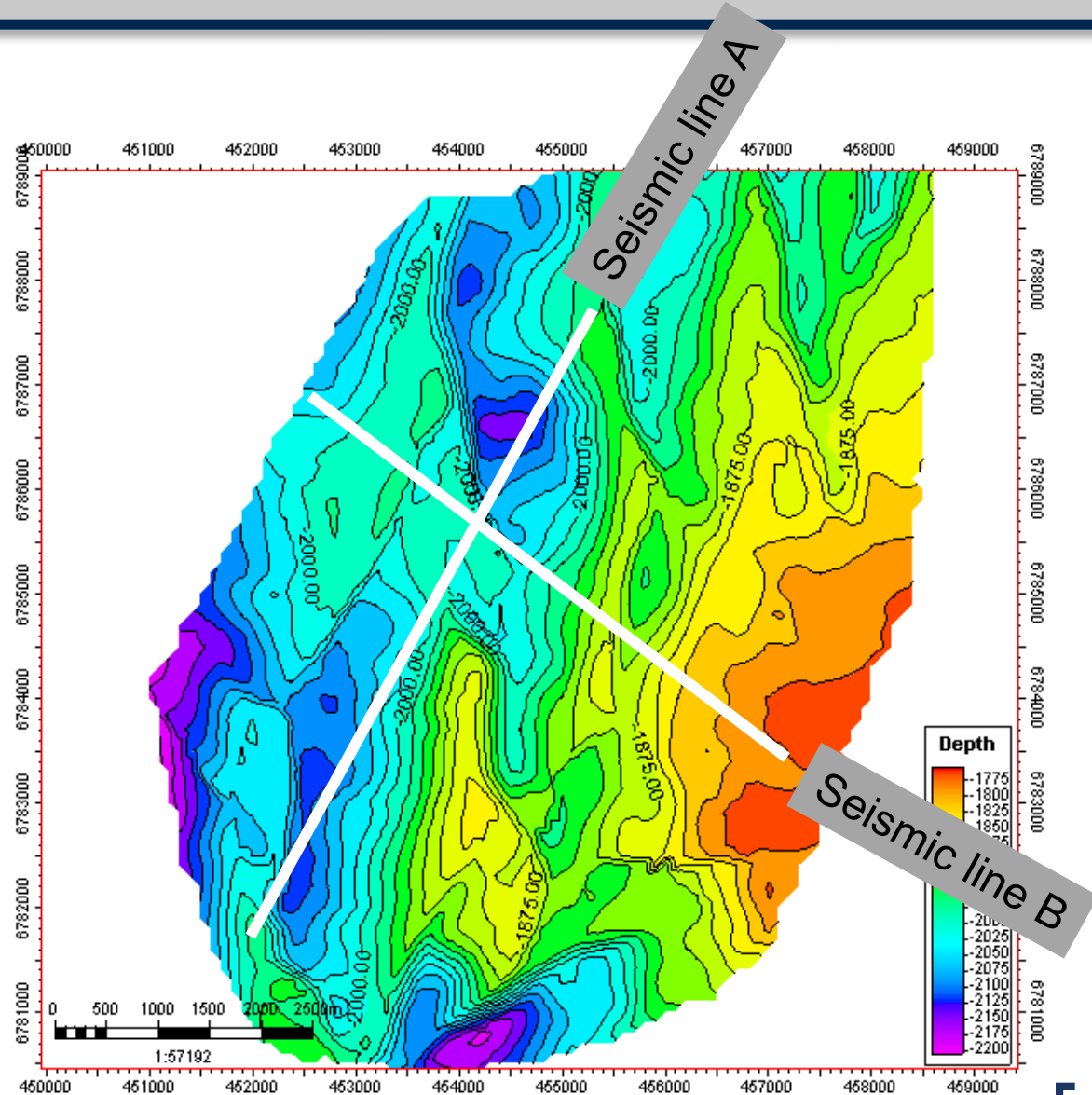
- Seismic interpretation and mapping techniques were revolutionized with the introduction of sequence stratigraphy and 3-D data sets
- The basic principles determine the ultimate success of the interpretation and mapping project
- Review of the basic principles of seismic interpretation and integration of seismic data with well log data
- Define the fundamentals of seismic reflection acquisition
- Define the fundamentals of seismic reflection responses at the borehole-synthetics

Seismic Data applied to Subsurface Interpretations

- Two major benefits of seismic data
 - Can be acquired in frontier areas beyond existing control
 - I.e., interpretations or maps can be extended into areas with no control
 - Provides explicit 2-D and 3-D control
 - As opposed to 1-D well data
- Very important that seismic interpreters understand geophysical principles

Caveat with Seismic Data

- The most insidious and potentially dangerous pitfall is assuming that all the data on a seismic section represents a planar slice through Earth **directly underneath** the line
- The solid lines show the surface locations of seismic lines A and B
- Subsurface reflections do not occur vertically beneath the surface locations when the reflecting horizons are dipping



Assumptions and Limitations

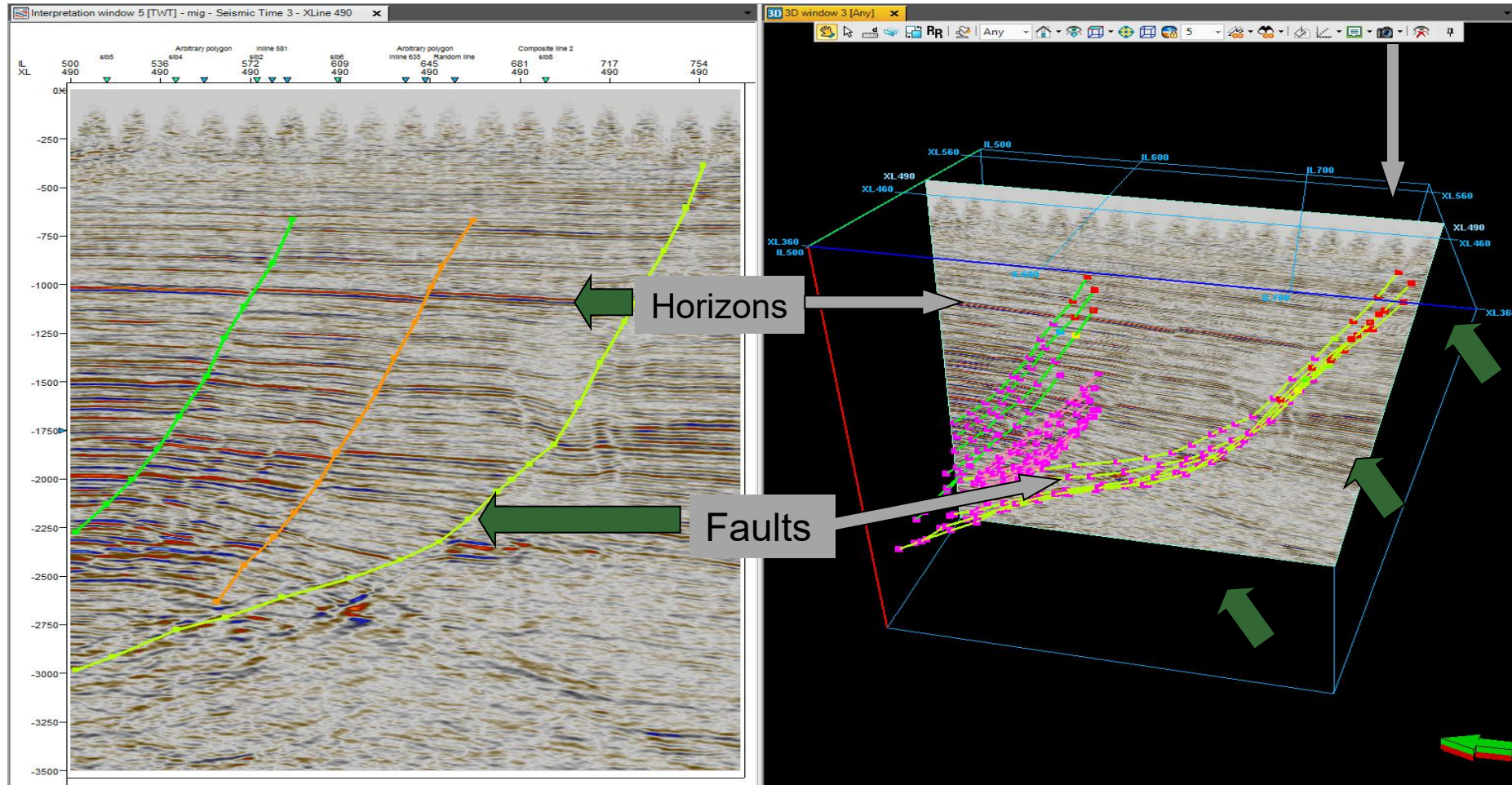
- For our purposes, we'll assume the following:
 - Seismic data has been properly acquired and processed (migrated and stacked properly)
 - Subsurface stratigraphy permits good data quality
- Our use of the seismic data to interpret subsurface is limited by:
 - Incomplete sampling even with 3-D data
 - Incomplete understanding of velocity variations
 - Limited resolution of the seismic wavelet with respect to geologic layering (vertical resolution)

The process for making subsurface maps from seismic

- Data validation (QC)
 - Are reflections related to geology?
- Geologic interpretation
 - correlation between traces
- Extracting information from seismic
 - Posting
 - Seismic and well log info
- Construction of subsurface maps
 - “Culmination” of all work
- Iteration and review
 - Double checks
 - Incorporation of new data

Keep in mind that additional data from either drilling or seismic will almost always change some of your ideas...

The process for making subsurface maps from seismic



Many seismic lines are shown with a vertical axis in time, not depth

First, we need to decide what the data represents – Horizons? Faults?

NEXT

Issues that we might face when using Seismic

- Severe horizontal velocity gradients
- High noise areas
- High bed dip
- Extremely complex geology

These characteristics of the subsurface may invalidate our assumptions.

Seismic data interpretation should be guided by the geology!

Reflections may or may not represent all discrete sedimentary boundaries

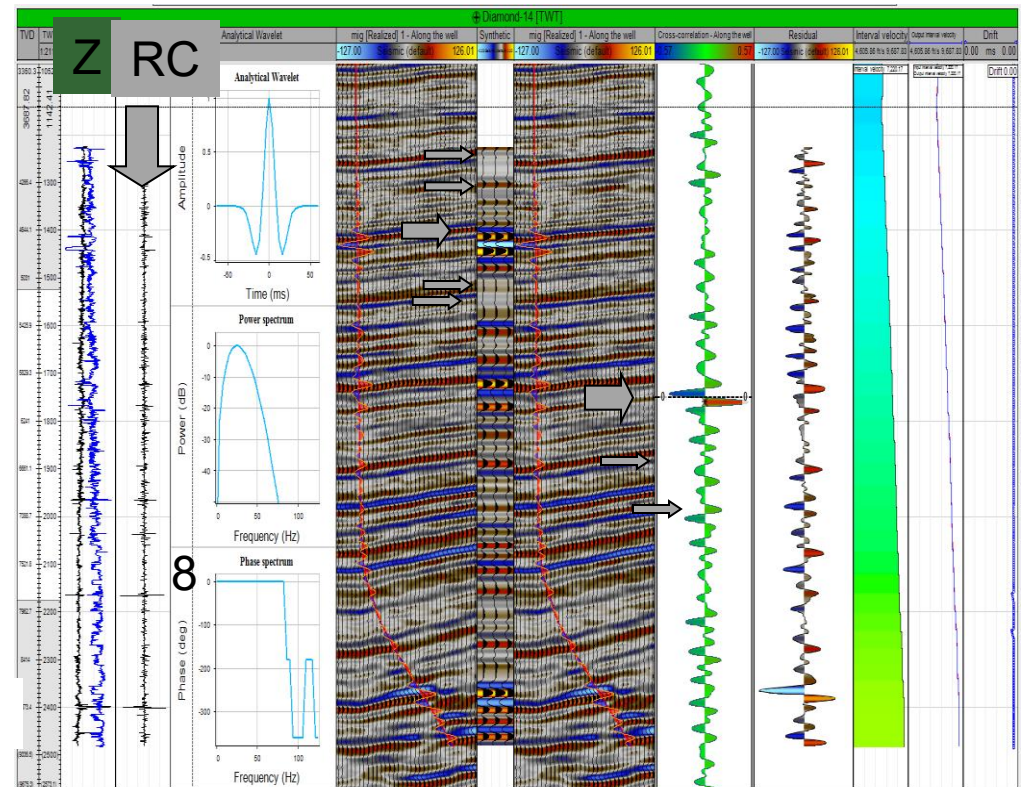
- The vertical pattern of reflections on seismic depends upon:
 - The actual stratigraphy
 - The frequency content of the processed seismic signal

Reflections on a seismic trace aren't geology... they're a composite of the seismic wave and the reflection coefficients.

Acoustic Impedance (Z) = density * velocity
Reflection Coefficient = $(Z_2 - Z_1) / (Z_2 + Z_1)$

Here's a lithology log showing 8 sedimentary units.

Seismic waves reflect at depths where the AI (Z) changes.



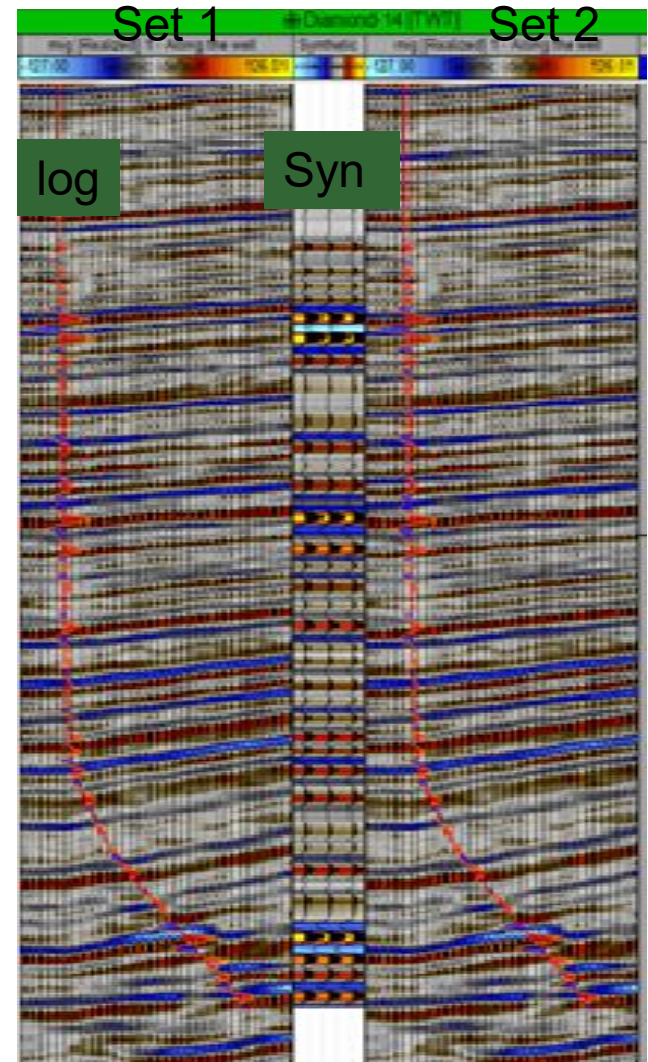
Tie Logs to Seismic – the Synthetic Seismogram.

The synthetic is based on the physical properties determined in well logs (velocity and density).

Notice how the size of the seismic wiggles is far larger than the spacing of variations indicated on the log.

I.e., the resolution of the seismic wavelet is coarse with respect to the bedding of sedimentary layers at this well bore.

Conclusion : we should not expect seismic to perfectly record all subsurface reflections.



Seismic Section Nomenclature

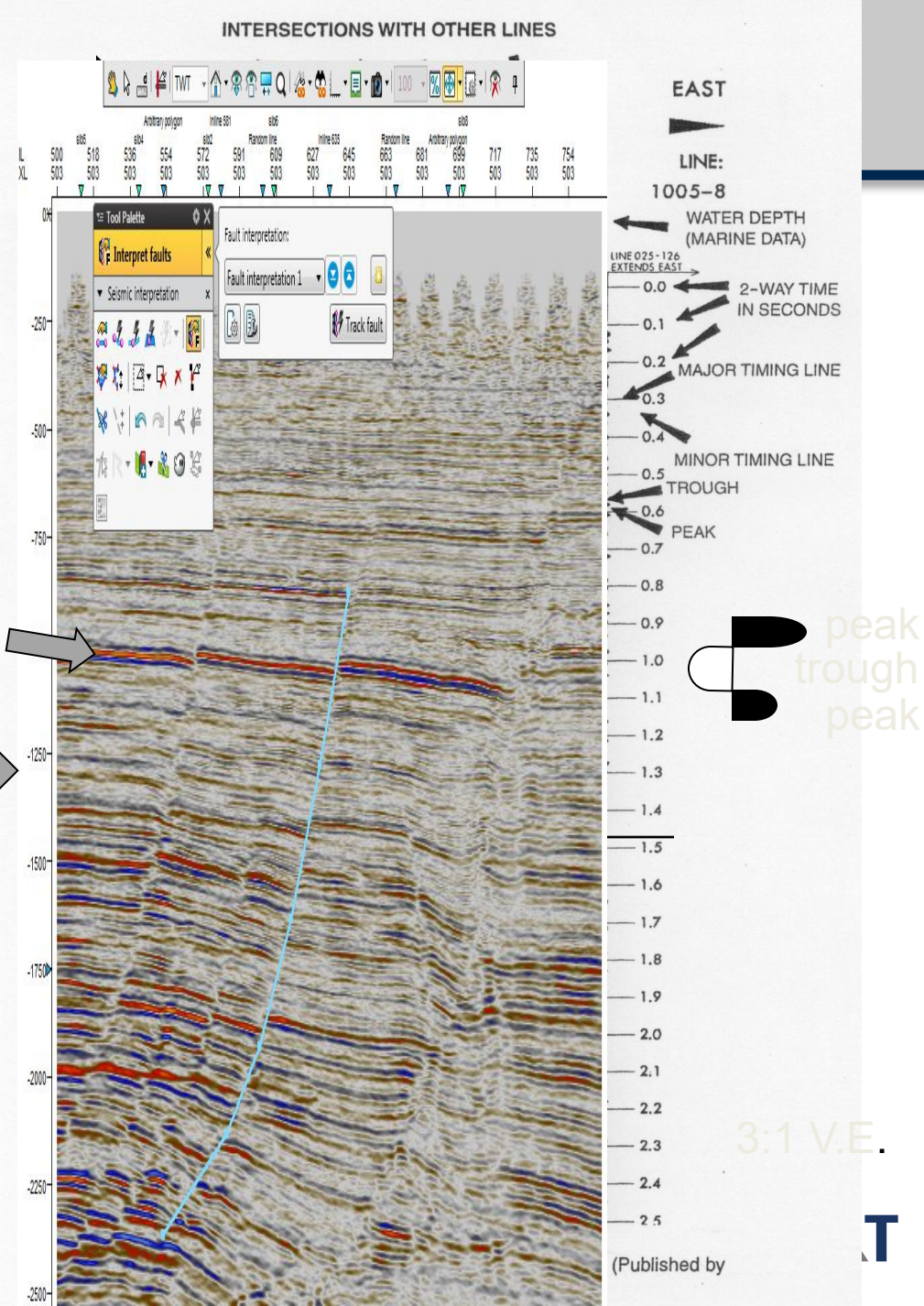
Line number
Shotpoint number

Troughs and peaks of seismic traces

Time lines on vertical scale

Vertical exaggeration e.g. 3:1

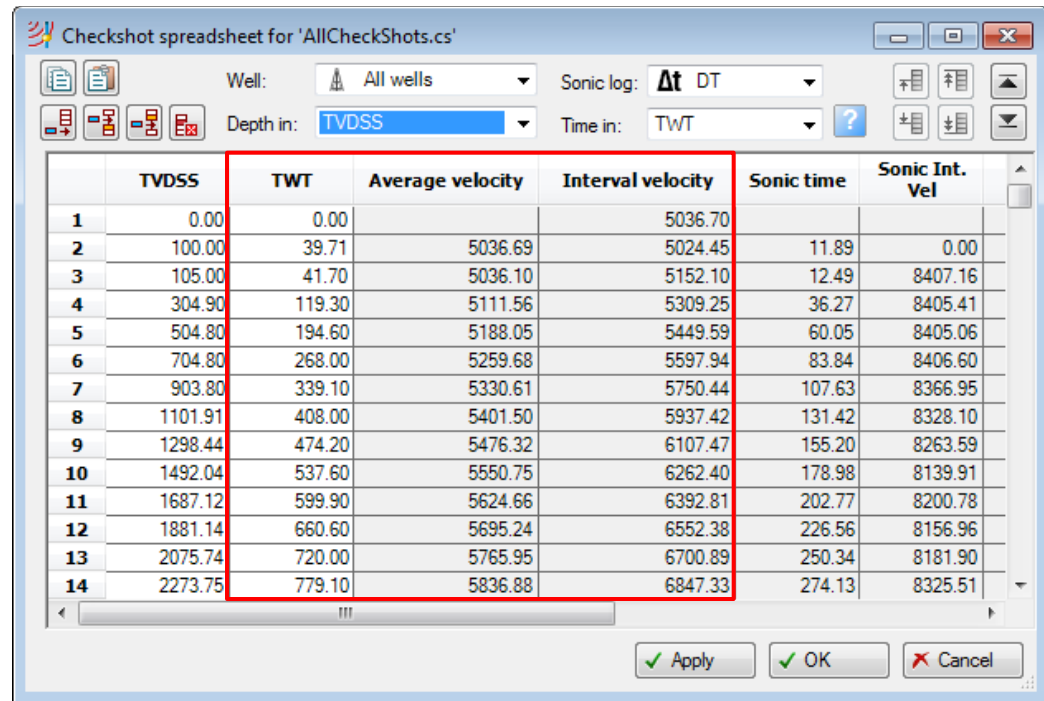
Intersection with other seismic lines



Getting a Time-Depth Relationship - Checkshots

Correctly loaded checkshot survey parameters – TWT, TVDSS, and velocities – are positive numbers.

The survey is conducted by lowering a string Geophones (pressure sensitive recorders) into a borehole and measuring the time it takes for a seismic wave to travel down to the receiver. This is then matched with the known depth of the recorders.



Checkshot spreadsheet for 'AllCheckShots.cs'

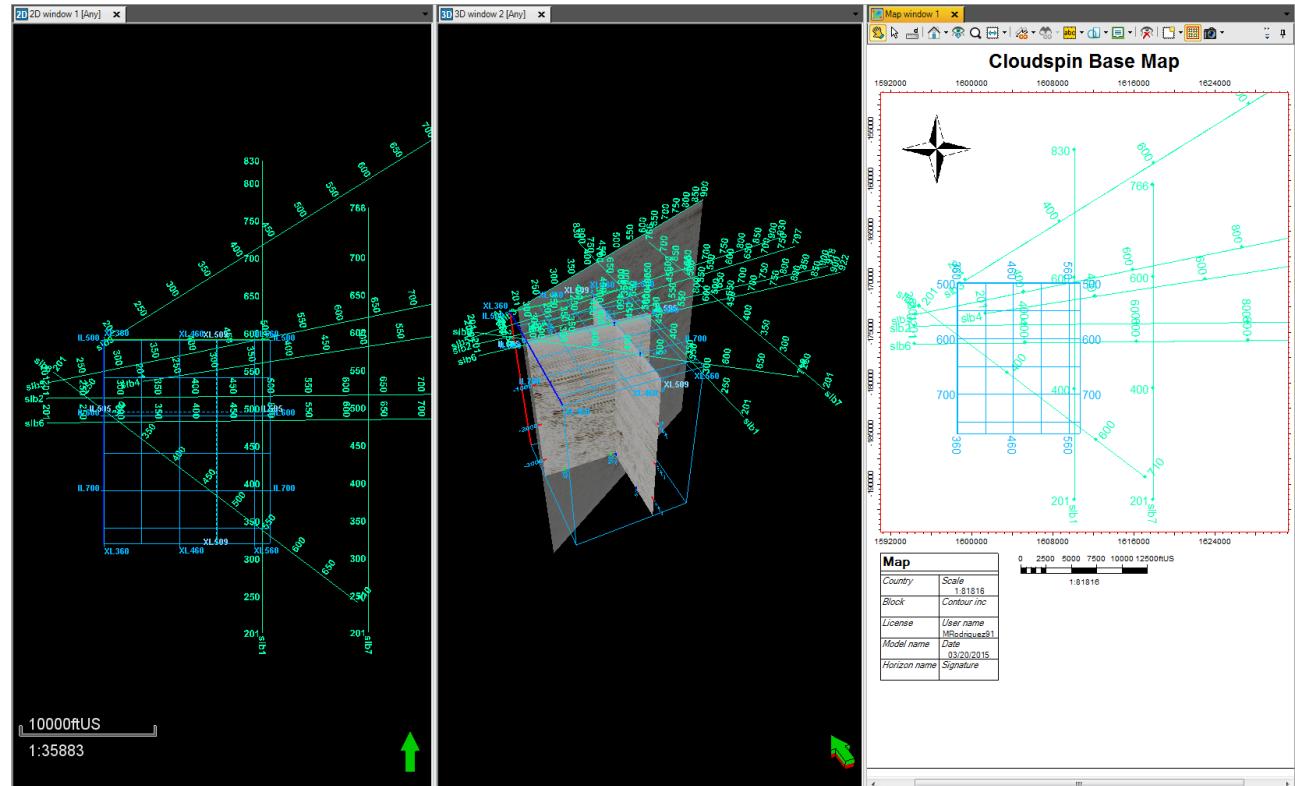
Well: All wells Sonic log: Δt DT
Depth in: TVDSS Time in: TWT

	TVDSS	TWT	Average velocity	Interval velocity	Sonic time	Sonic Int. Vel
1	0.00	0.00		5036.70		
2	100.00	39.71	5036.69	5024.45	11.89	0.00
3	105.00	41.70	5036.10	5152.10	12.49	8407.16
4	304.90	119.30	5111.56	5309.25	36.27	8405.41
5	504.80	194.60	5188.05	5449.59	60.05	8405.06
6	704.80	268.00	5259.68	5597.94	83.84	8406.60
7	903.80	339.10	5330.61	5750.44	107.63	8366.95
8	1101.91	408.00	5401.50	5937.42	131.42	8328.10
9	1298.44	474.20	5476.32	6107.47	155.20	8263.59
10	1492.04	537.60	5550.75	6262.40	178.98	8139.91
11	1687.12	599.90	5624.66	6392.81	202.77	8200.78
12	1881.14	660.60	5695.24	6552.38	226.56	8156.96
13	2075.74	720.00	5765.95	6700.89	250.34	8181.90
14	2273.75	779.10	5836.88	6847.33	274.13	8325.51

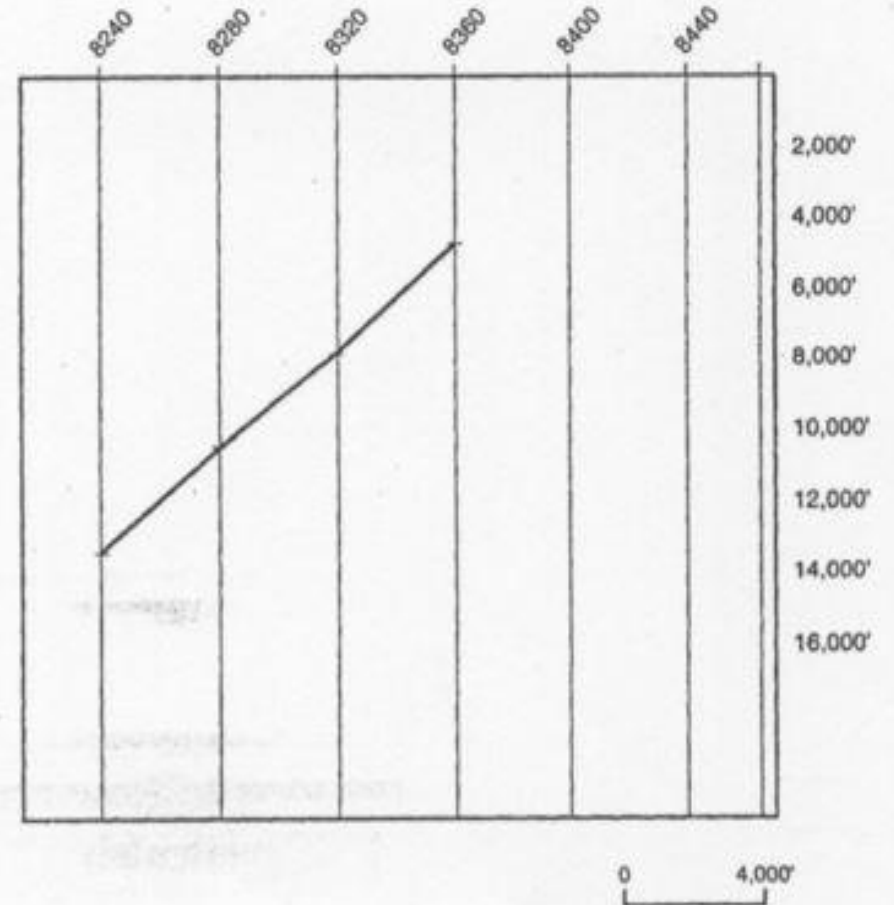
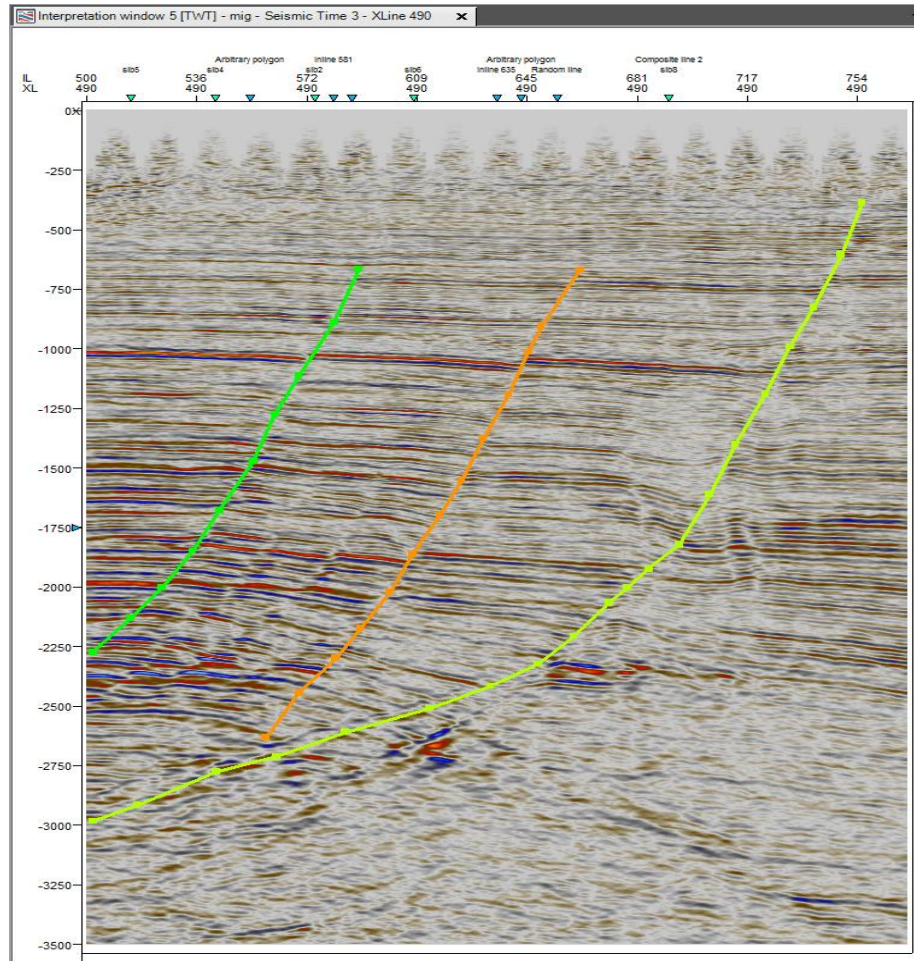
Apply OK Cancel

Base Maps

- Base maps are important aids during seismic interpretation.
- Seismic interpreters frequently navigate through base maps during their interpretation.
- You can display new seismic acquisition plans in the base map as culture data.



Is the Fault Really Curved in Depth?



Our fault from the previous slide appears curved in time (left)
In reality, it is straight (right)

Orientation of the Seismic Line creates Apparent Structure.

E.g., this fault appears to be listric along seismic Line A

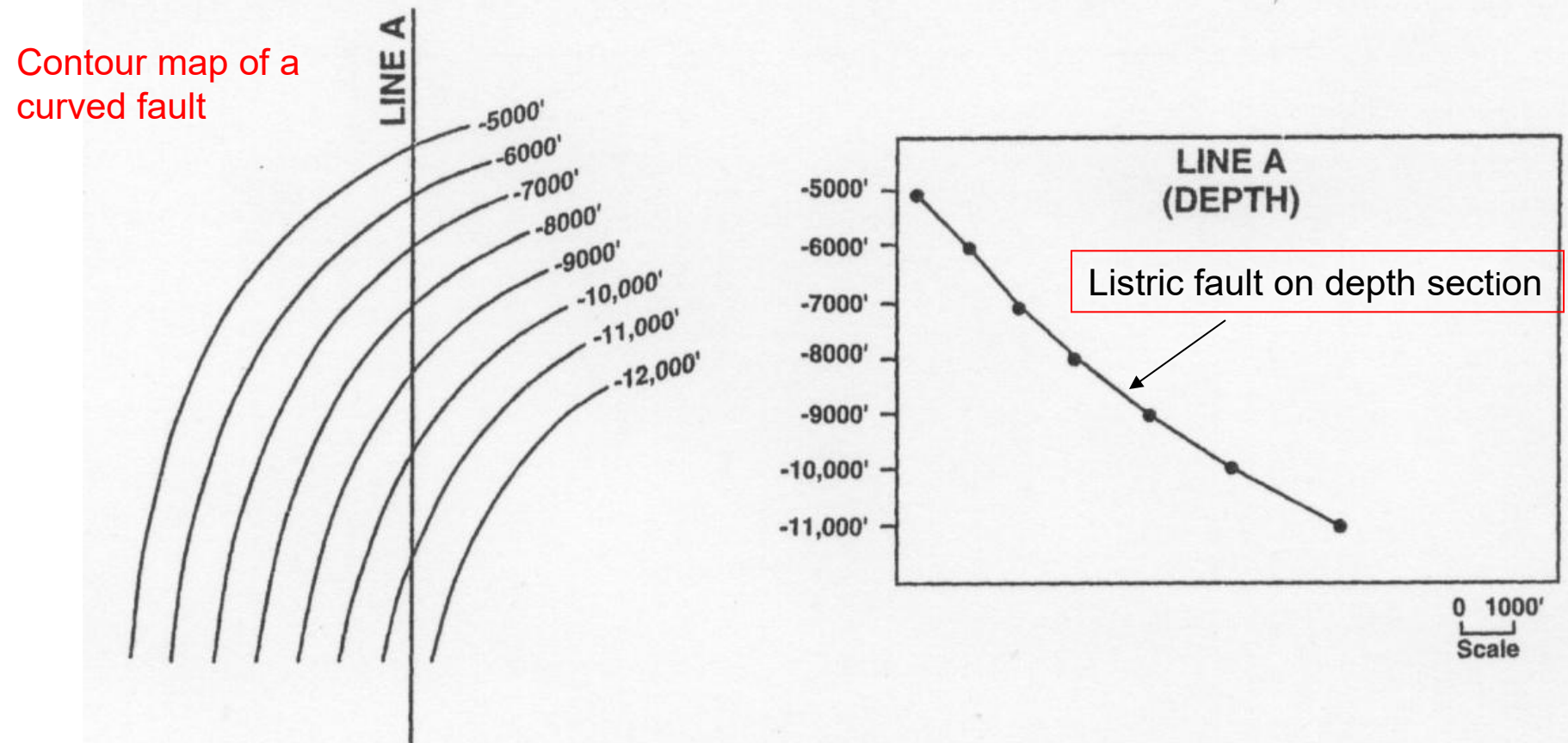


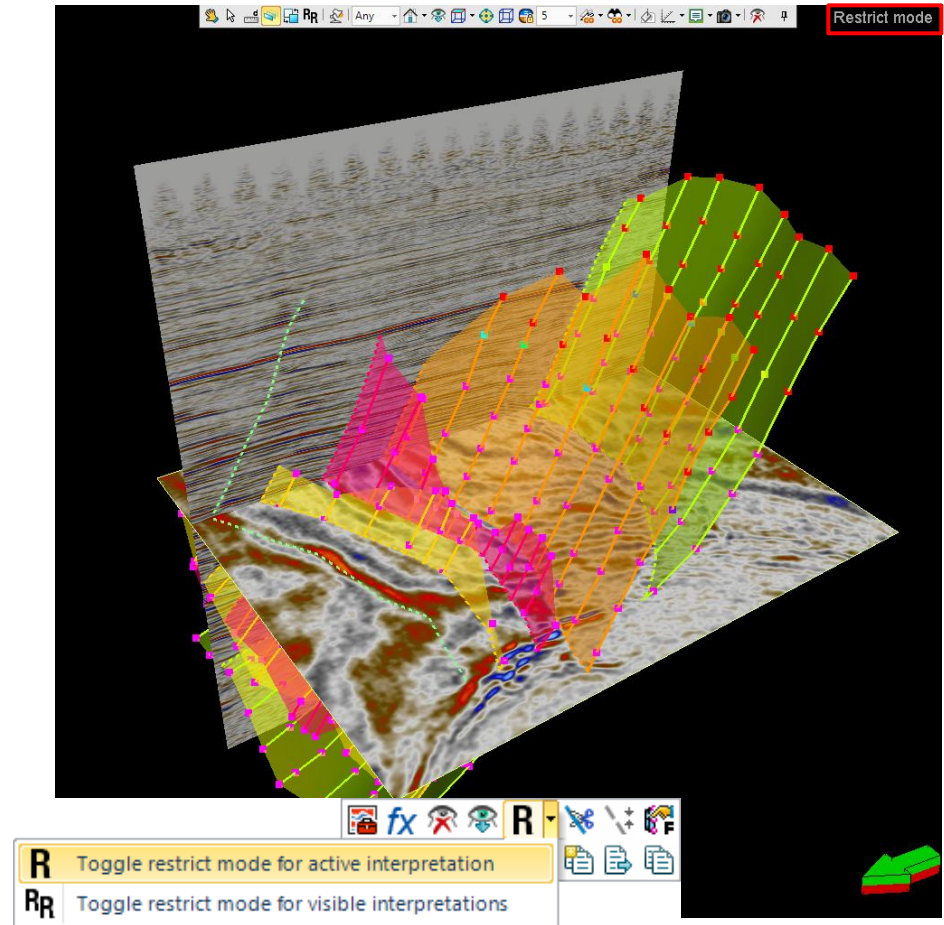
Figure 5-10 Line orientation causing an apparent listric fault on seismic line. The dip on the fault plane is uniform; however, the line orientation makes the fault appear listric, which it is not.

Tying 2D line to seismic and horizon interpretation

How do we use 2D data to develop a 3D view of subsurface horizons?

1. Display interpretations and seismic lines in a **3D window**.
2. Right-click one fault and in the mini toolbar, click *Restrict mode* and click *Toggle restrict mode for active interpretation*.

By tying the data, we ensure that the same event (or map horizon) are at the same subsurface elevation where the two lines intersect.



Tying seismic data

In the upper left and right are two possible interpretations of faulting from correlations between the wells 1, 2, 3, and 4.

Both fault interpretations respect the data in the 4 wells.

Tying the faults on the two seismic lines shown on the middle figures can help eliminate some ambiguity in interpretation

Figure D shows the intersection of lines A and B with the interpretation of Faults Y and X posted.

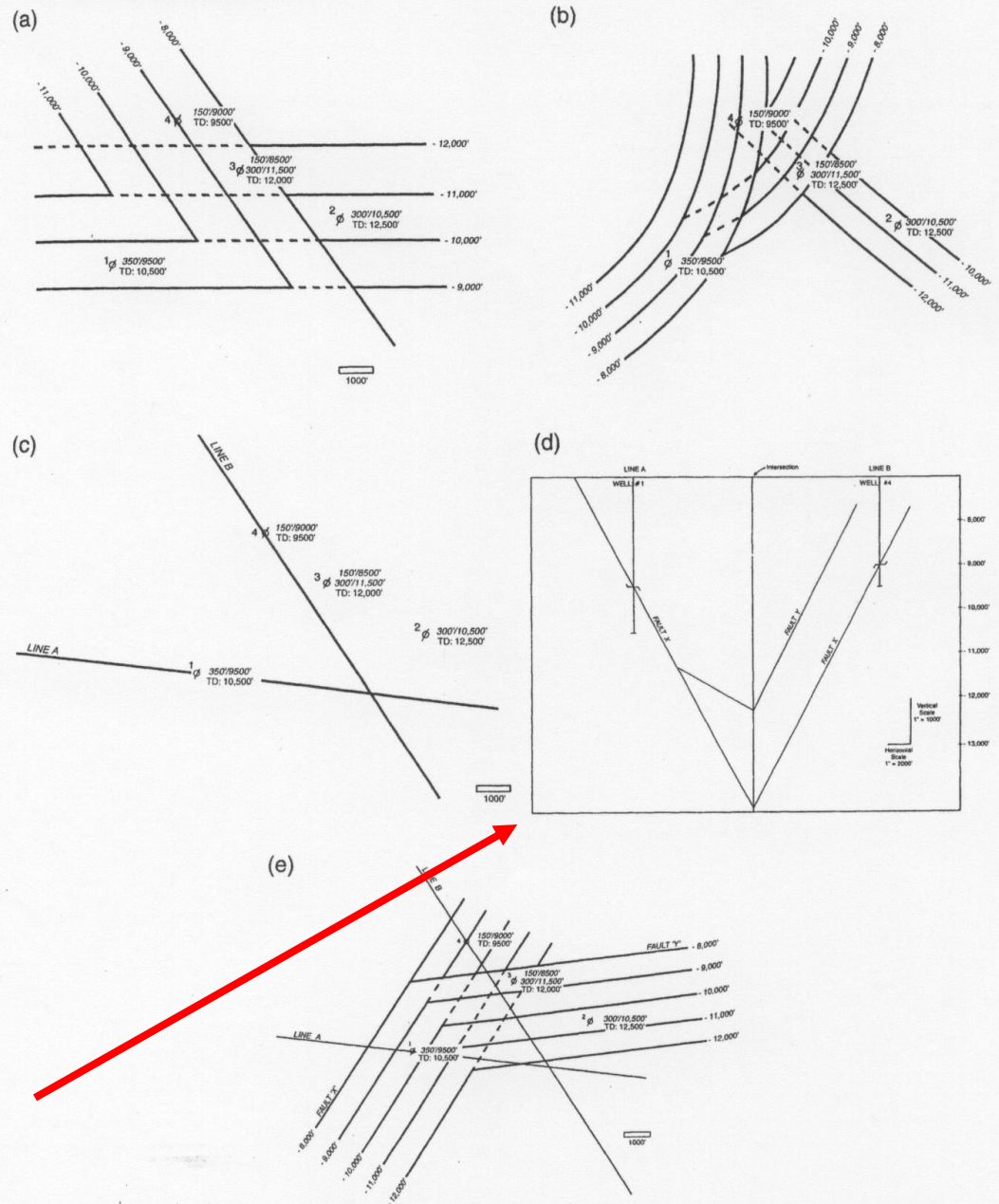


Figure 5-12 (a) Possible interpretation of hypothetical fault data observed in a set of wells. (b) Another possible interpretation of the same fault data. Multiple fault surface interpretations are possible from this set of data. (c) Hypothetical seismic grid through faulted area. (d) Appearance of hypothetical seismic sections when interpreted and tied. Note that fault traces on profiles meet (tie) at the intersection of the two lines. (e) Final fault surface interpretation, integrating both seismic and well data.

Tying seismic data

How can you resolve where the 2 faults on dip line X correlate to the set of 4 faults seen on dip line Z?

Unfortunately, the faults are poorly imaged on strike Line Y

One reasonable first guess would be to assume that faults that buried at the same depth are the same.

More closely spaced dip lines between X and Z might help resolve the ambiguity.

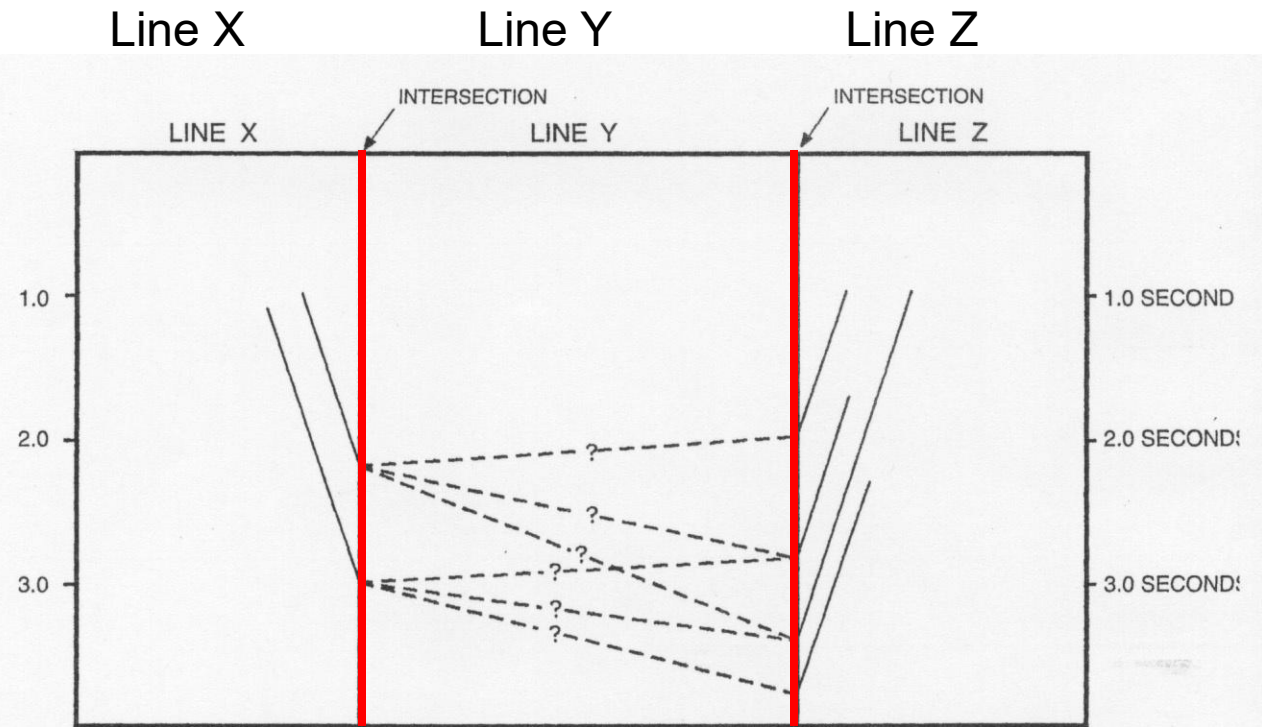
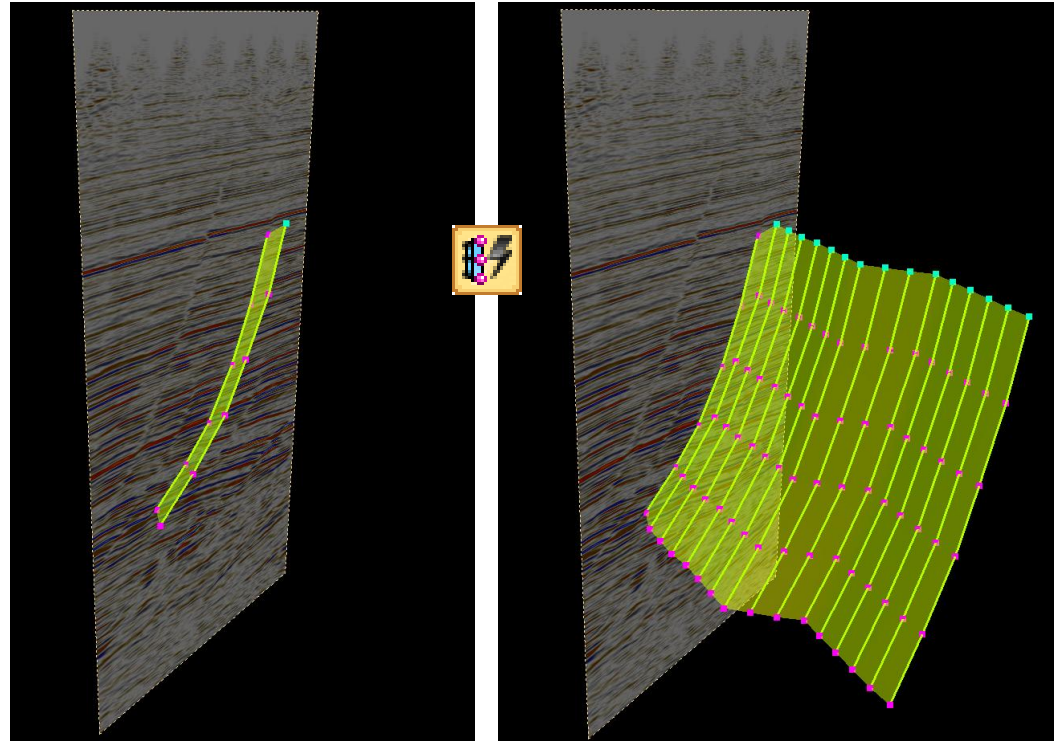


Figure 5-13 Ambiguity when tying closely spaced faults. Line Y is a strike line. Multiple interpretations may be possible.

Autotracking of Faults in Petrel

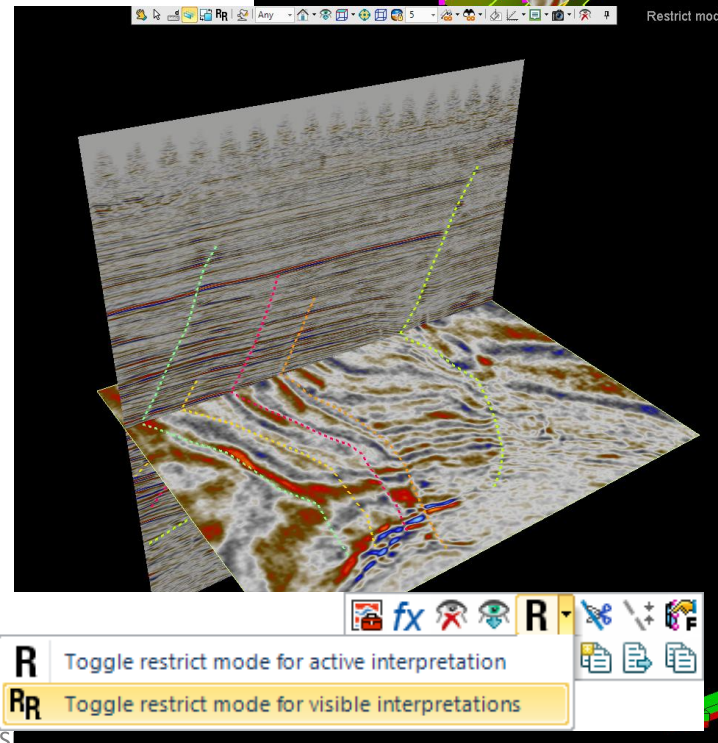
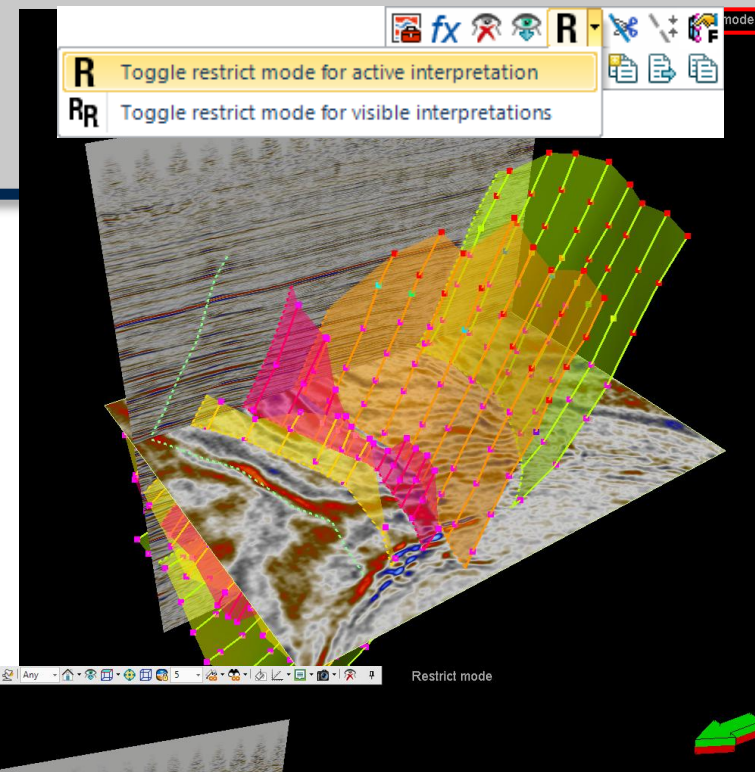
1. Create a new fault interpretation and interpret it manually on one or two seismic lines. Adjust settings on the **Autotracking** tab in the **Settings** dialog box.
2. Right-click the fault and click *Track faults*  in the mini toolbar.
 - The fault extends along the seismic cube.



Restrict Mode

1. Display interpretations and seismic lines in a **3D window**.
2. Right-click one fault and in the mini toolbar, click *Restrict mode* and click *Toggle restrict mode for active interpretation*.
3. Select *Toggle restrict mode for visible interpretations*, which restricts the display to all displayed interpretations.

To undo, click **RR** in the **Window** toolbar.



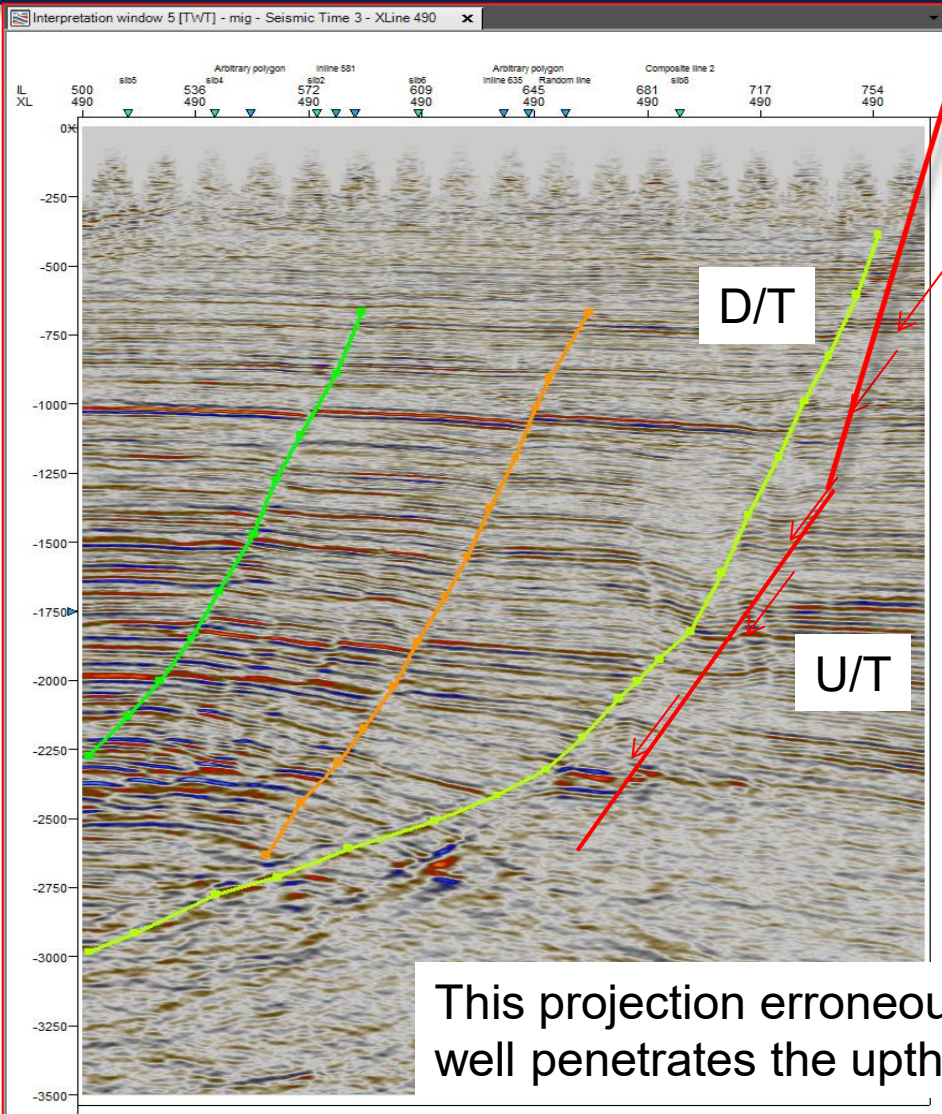
Viewing Seismic in Petrel

- Arrange sections with similar orientations
- Begin with seismic section that images the dip direction of the predominant geologic features
- Compare/contrast features on the next closest dip line to note what changes (question the data)
- Continue the process two lines at a time through all dip sections
- Follow the same procedure through all strike sections
- Map horizons should have good reflection strength and continuity
- Sequence boundaries are typically good reflections corresponding to isochronous surfaces

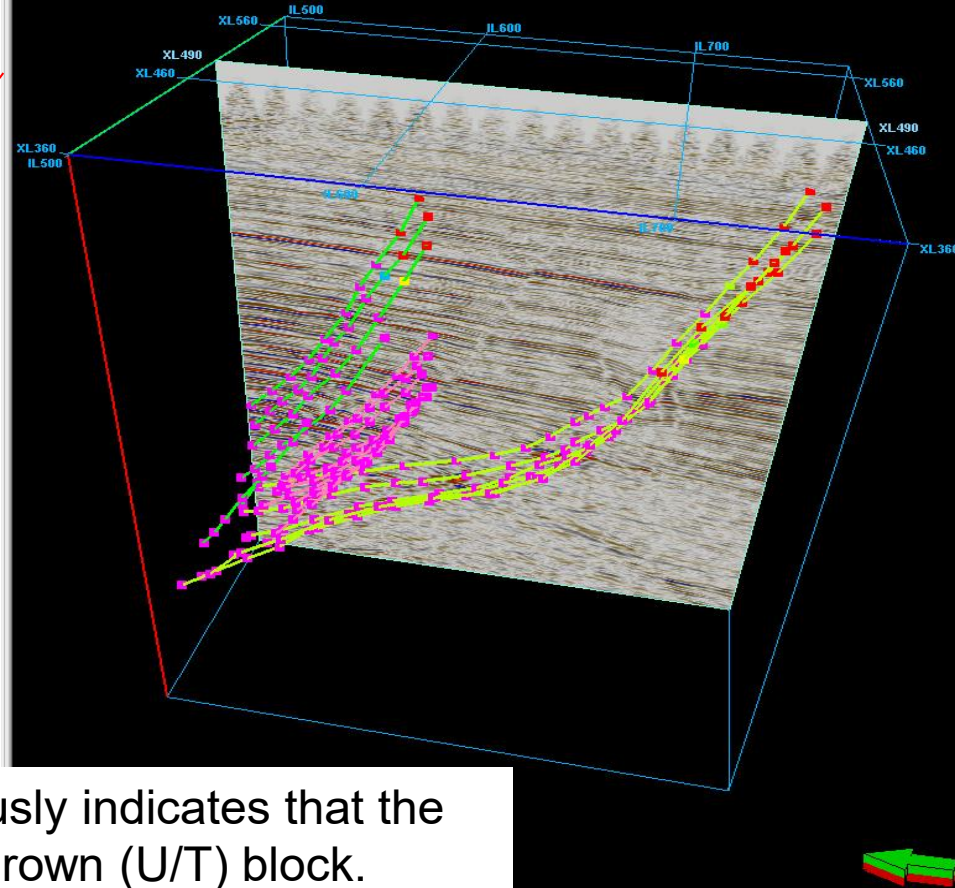
Annotating well information

- Post wells that intersect the seismic lines
- Tops and bases of units seen in well bores are posted on seismic
- Straight hole wells on a seismic line require only a check-shot survey
- Directional wells require a directional survey
- Wells not on a seismic line are usually projected to nearest seismic line
- Projections always pose a potential compromise

Annotating well information



Here's deviated well #2 projected to the nearest seismic line.



This projection erroneously indicates that the well penetrates the upthrown (U/T) block.

Annotating well information

Here's the surface location (SL) of well #2 with respect to the offset location of the seismic line.

The contours represent the fault on the RHS of the seismic section seen in the previous slide.

As you can see from the directional survey information at well #2 plus the regional mapping of the fault, the well remains above the fault at all depths. I.e., within the D/T block.

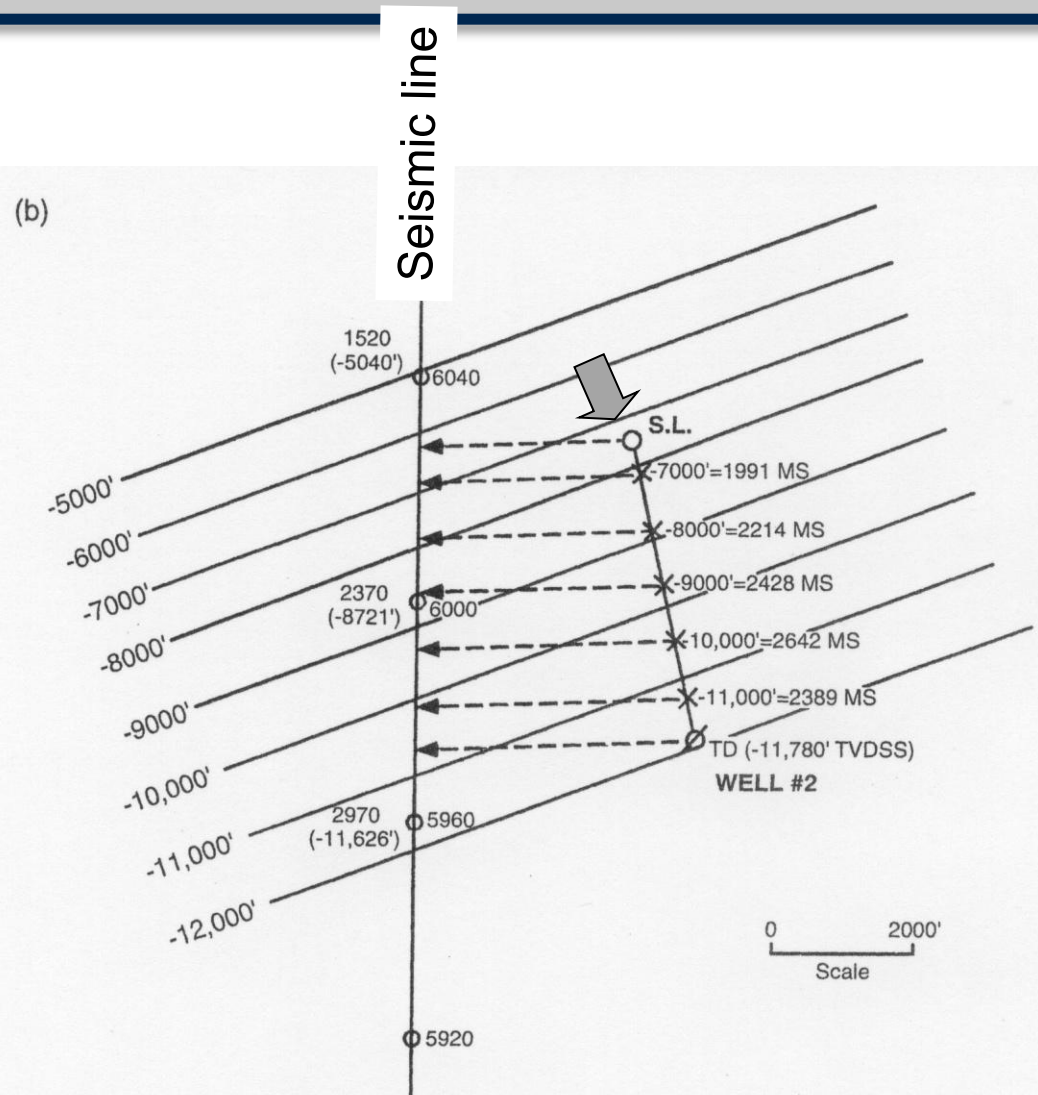


Figure 5-14b Fault surface map of fault observed in Fig. 5-14a. Projection onto line makes well appear to have penetrated the upthrown block. Fault surface and the well's directional survey clearly indicate that the well remained in the downthrown block. Care must be exercised in projecting wells onto seismic lines.

Tying the faults...

- Fault surfaces should be loop-tied prior to loop-tying horizons
 - In areas of complex structures, the fault style determines the map structure
 - In areas of variable data quality, some isolated lines may not show a fault.
 - I.e., the horizon will be projected without offset
 - The resulting mis-tie might ultimately require a non-existent fault to be inserted

Loop tying horizons...

- Post all intersections
- Start with largest and most obvious features to be mapped
- Tie to all seismic lines in survey area
- Next, concentrate on second order features
- Pick a loop-tying path that requires the least amount of assumptions
 - A path that initially crosses the fewest number of faults
 - A path that initially crosses small offset faults

Fault Interpretation in Petrel

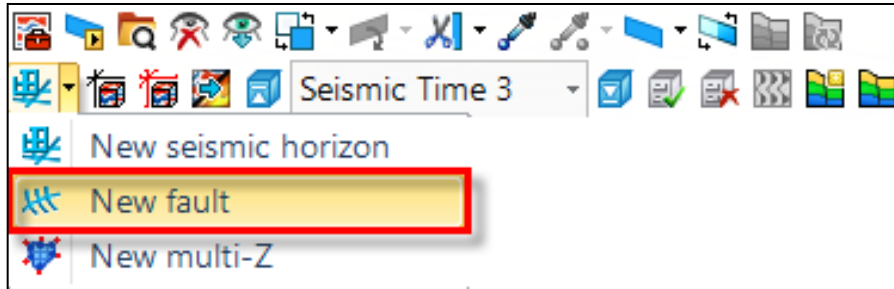
- Interpreting faults is one of the key tasks in any seismic interpretation workflow.
- You can interpret faults manually on inlines, crosslines, time slices, random lines, or arbitrary lines in 2D, 3D, and Interpretation windows.

In this module, you learn about:

- fault interpretation in **Petrel**
- manual fault picking
- manipulating and editing faults
- tracked faults
- use of **Interpretation manager**

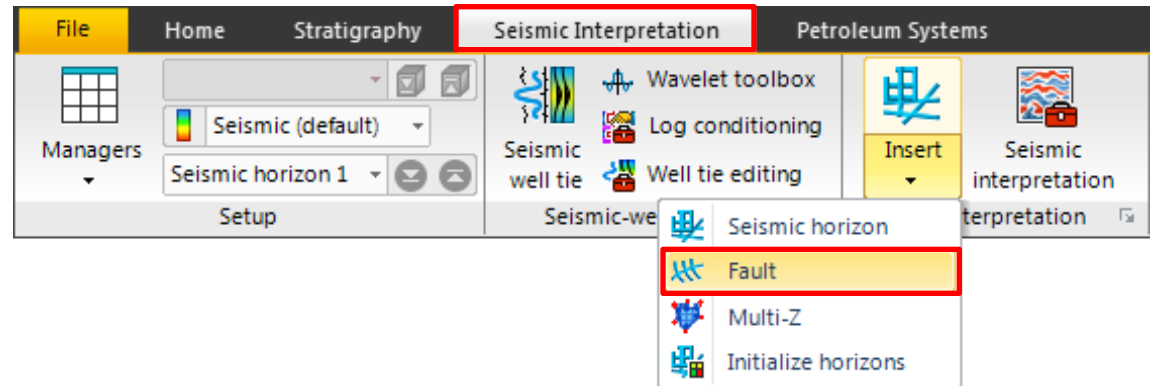
Create a new fault

- There are many ways to create or insert a new fault. Your choice depends on where you are in the application.



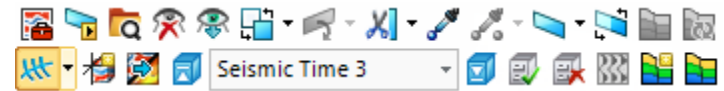
Mini toolbar

Seismic Interpretation tab



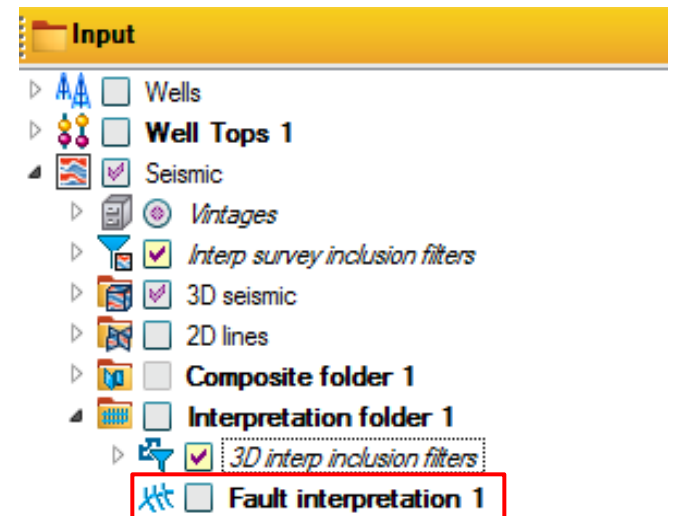
Manual fault picking (1)

1. Open a new **Interpretation** window and display an intersection from of a seismic cube.
2. Right-click the Seismic section to show the mini toolbar and click *New fault* in the mini toolbar.



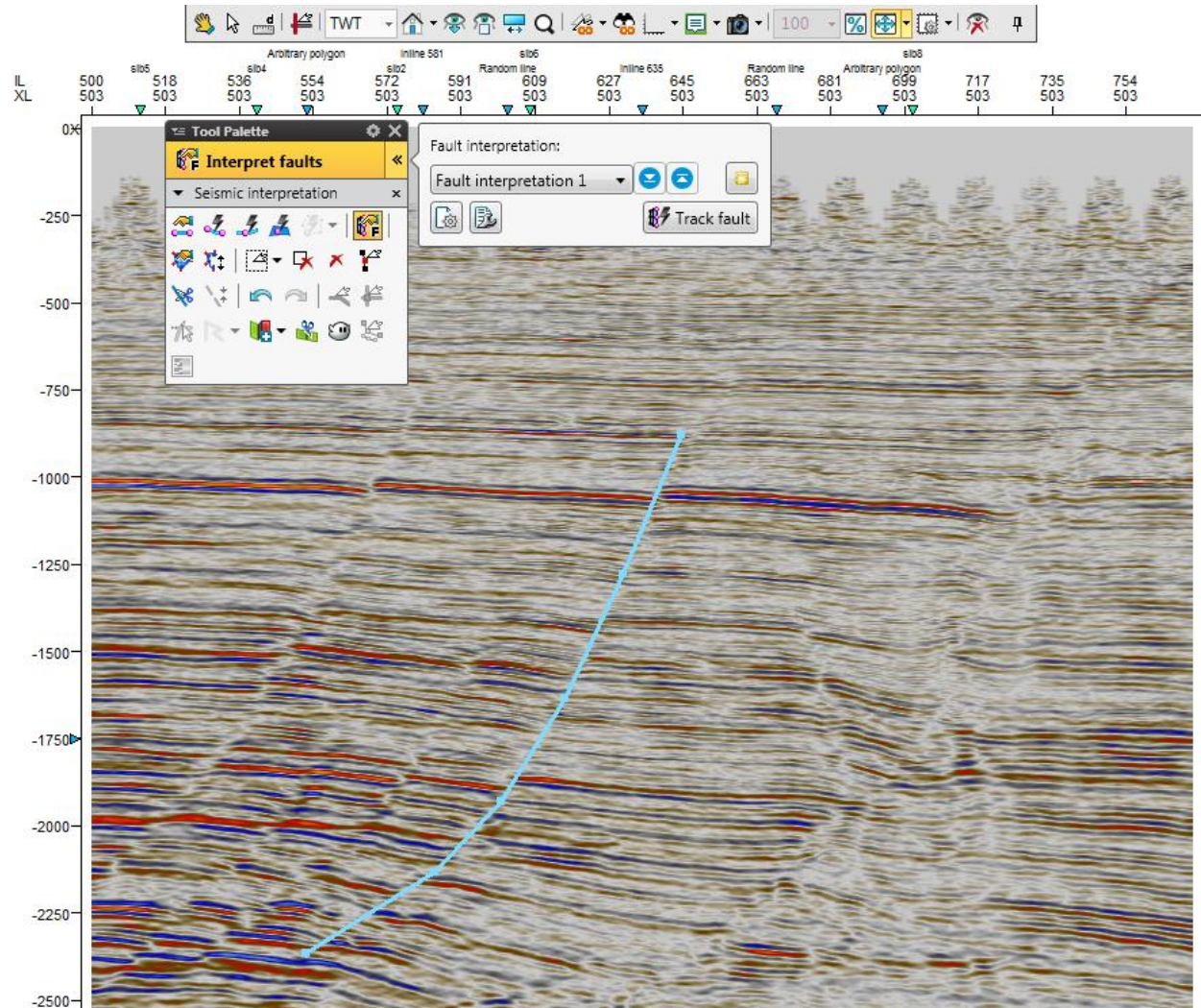
- A new fault is created in the **Interpretation** folder in the **Input** pane.

3. Click *Seismic interpretation Tool Palette* in the mini toolbar.



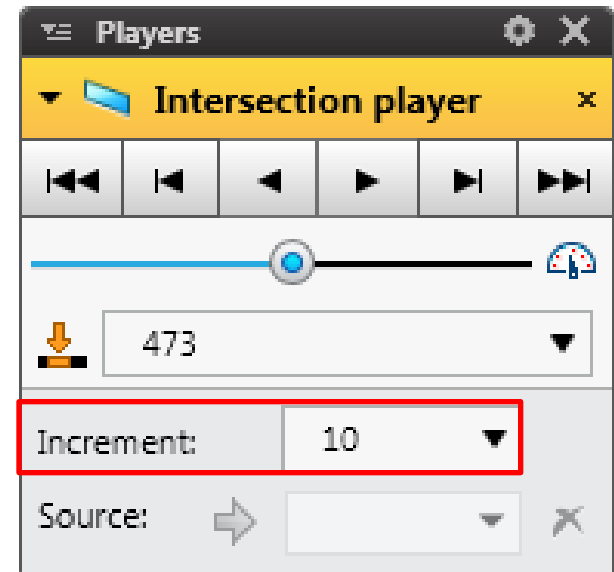
Manual fault picking (2)

4. After you insert a new fault, start digitizing the fault.
5. Double-click or press N to complete the fault segment.



Manual fault picking (3)

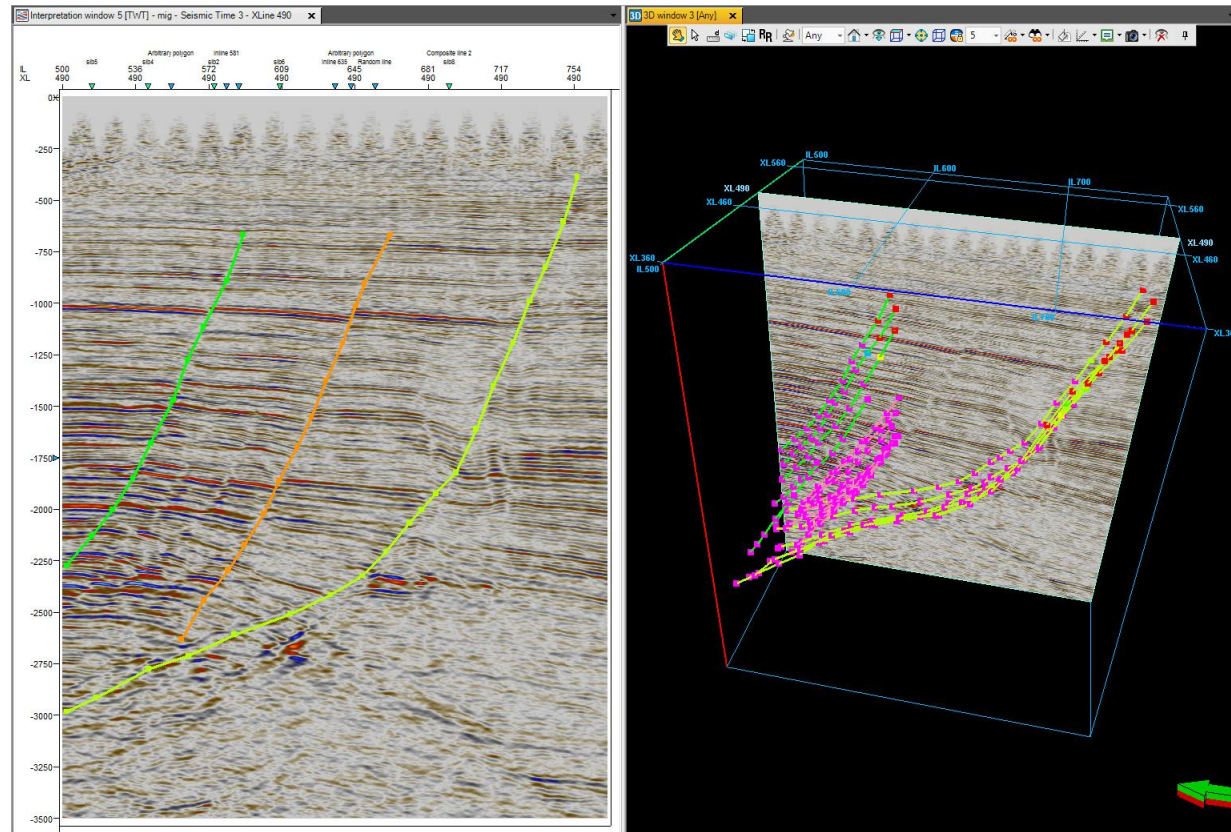
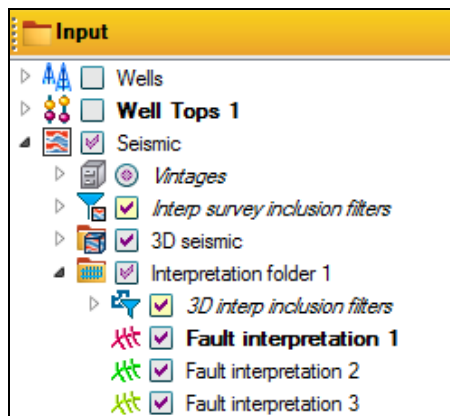
6. Click *Intersection player*  from the mini toolbar.
7. In the **Intersection player**, change the increment.
8. Activate the window, and press Page Up and Page Down to jump lines using the increment defined in your intersection player.
9. Click *Interpret faults*  in the mini toolbar or press F to continue the interpretation.



Manual fault picking (4)

10. Add as many new faults as you want, and digitize across the seismic data.

All inserted faults are saved in the default **Interpretation** folder in the Input panel.



Edit a fault segment (1)

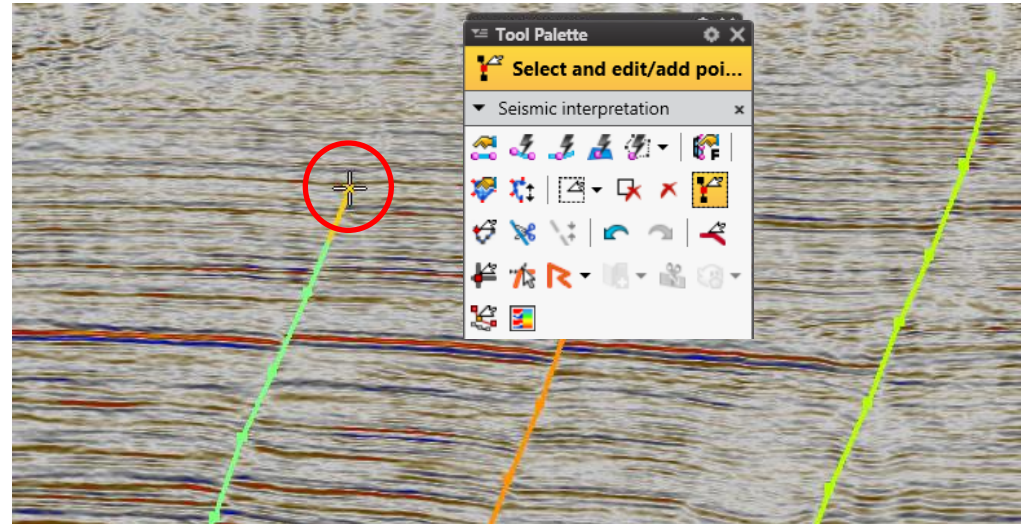
1. In the **Window** toolbar, click *Select [P]* mode and click the desired fault.



2. To edit a fault segment, go to the **Seismic interpretation Tool Palette** and click *Select and edit/add points*.

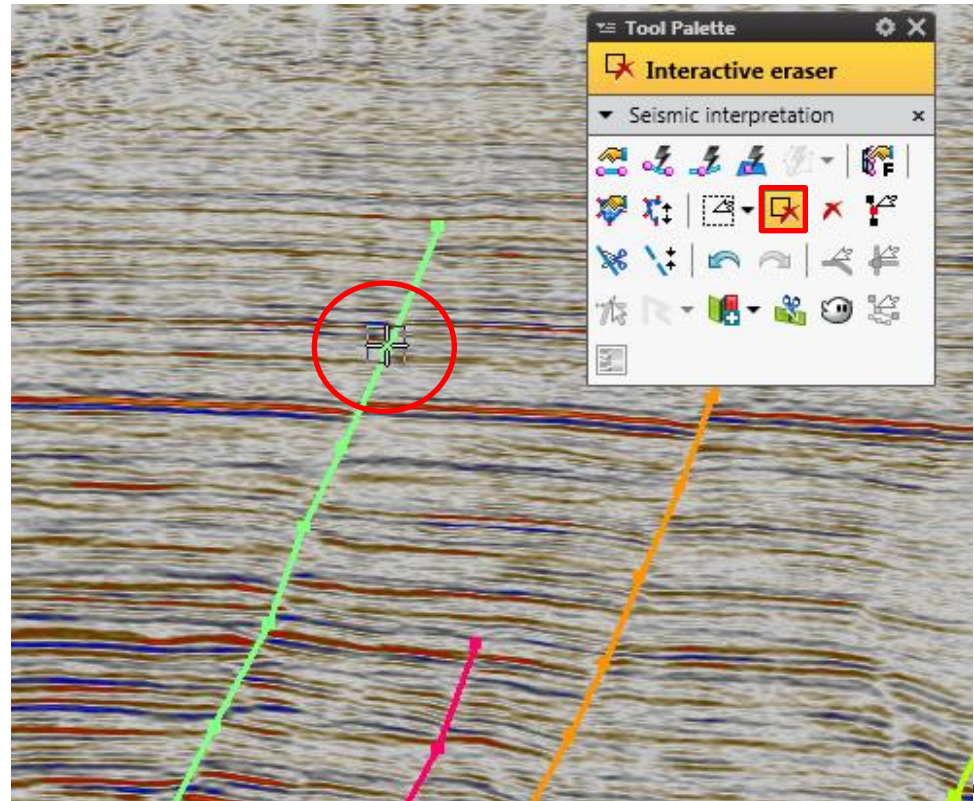


3. Move the points on the fault or add new points to the fault segment to accurately mark the fault geometry.



Edit a fault segment (2)

4. To delete all or part of a fault segment:
 - a. Click *Interactive eraser*  from the **Seismic interpretation Tool Palette**.
 - b. Click a digitized point on the active fault to remove it.

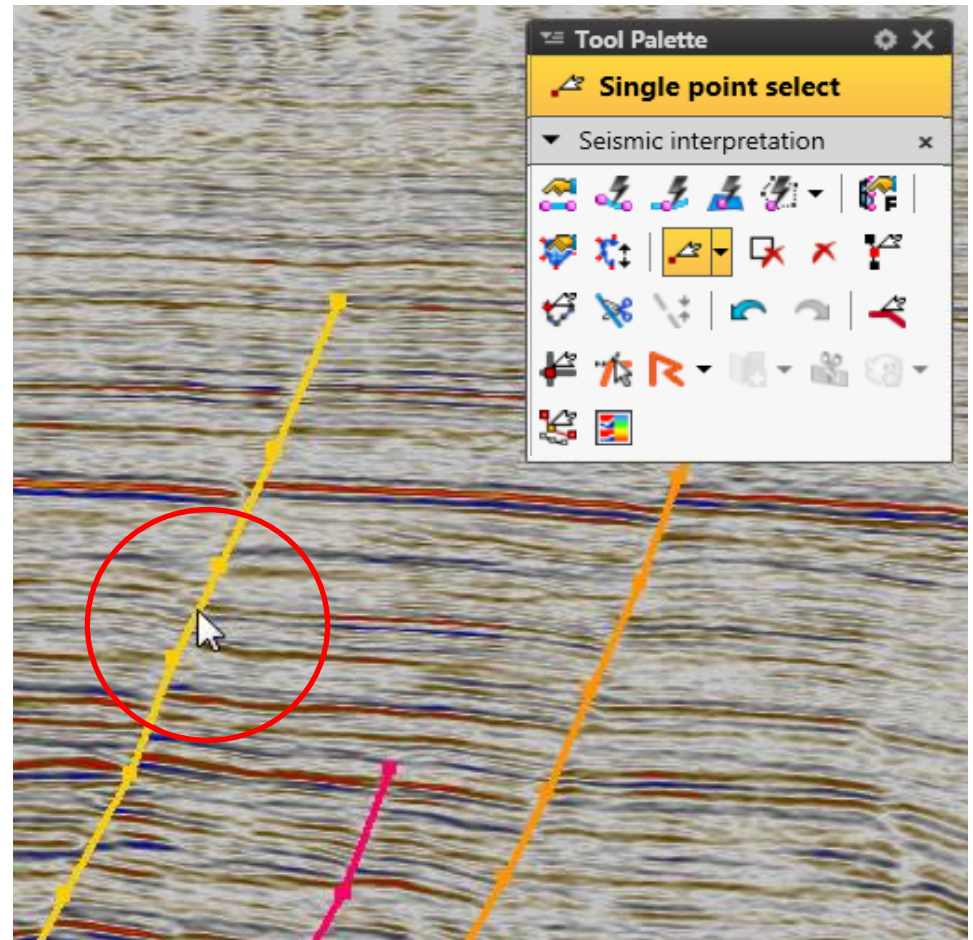


Move a fault segment (1)

1. Activate the fault.

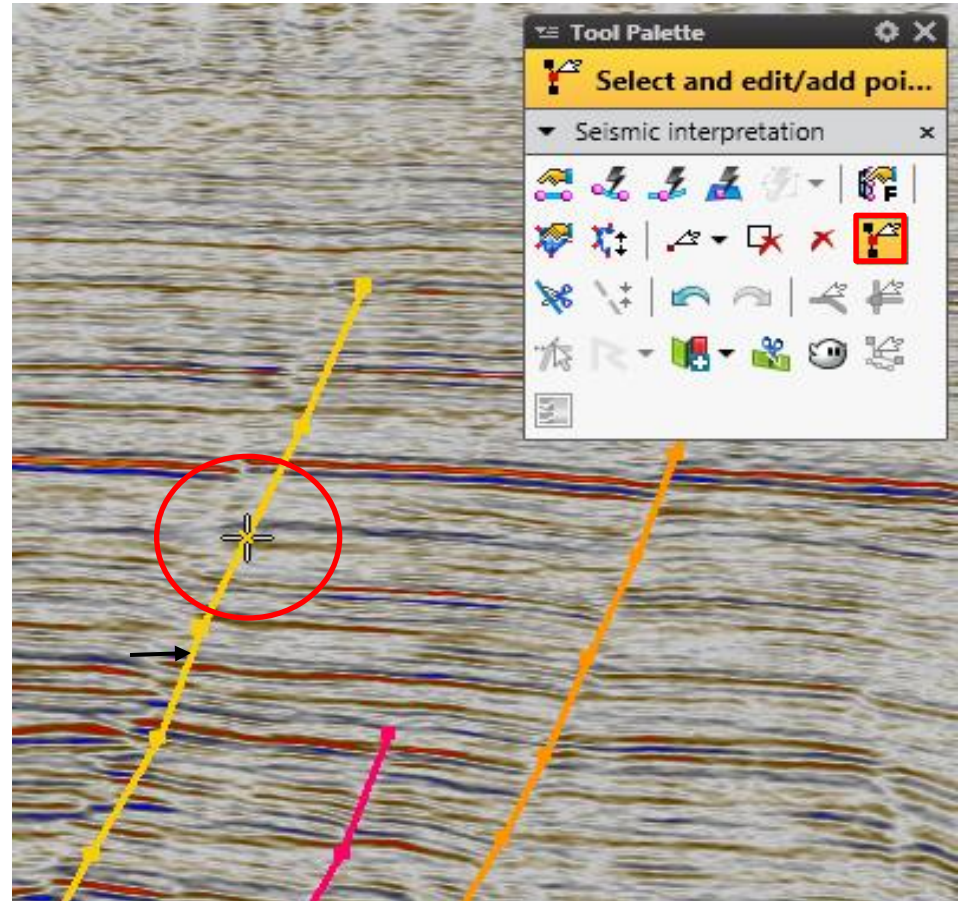
Click *Single point*  from the **Seismic interpretation Tool Palette**.

2. Click between two digitized points (nodes) on the fault segment to make the entire segment yellow.



Move a fault segment (2)

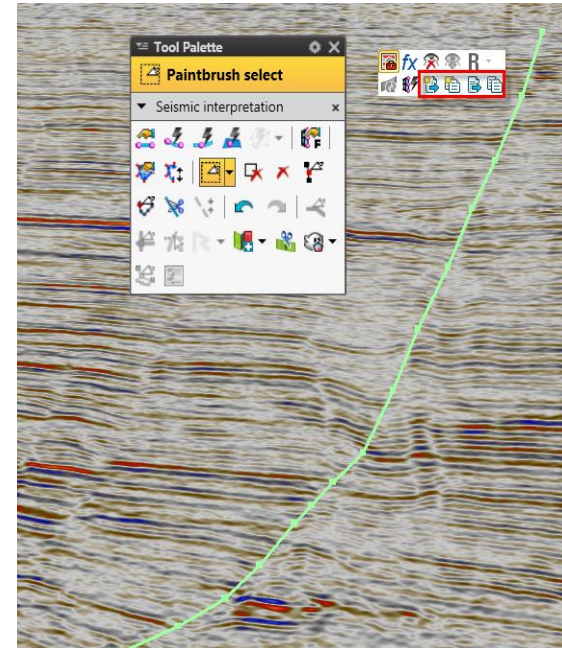
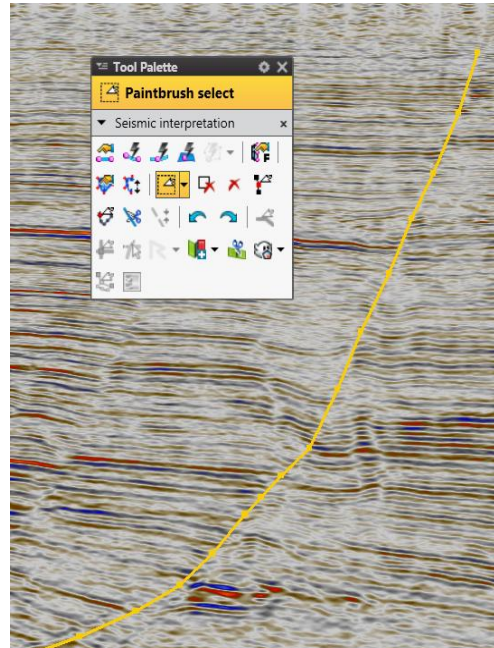
3. Click *Select and edit/add points*  on the **Seismic interpretation Tool Palette**.
4. Click a single digitized point (node) and drag it to move the whole segment to a new location on the seismic section.



Reassign fault segments

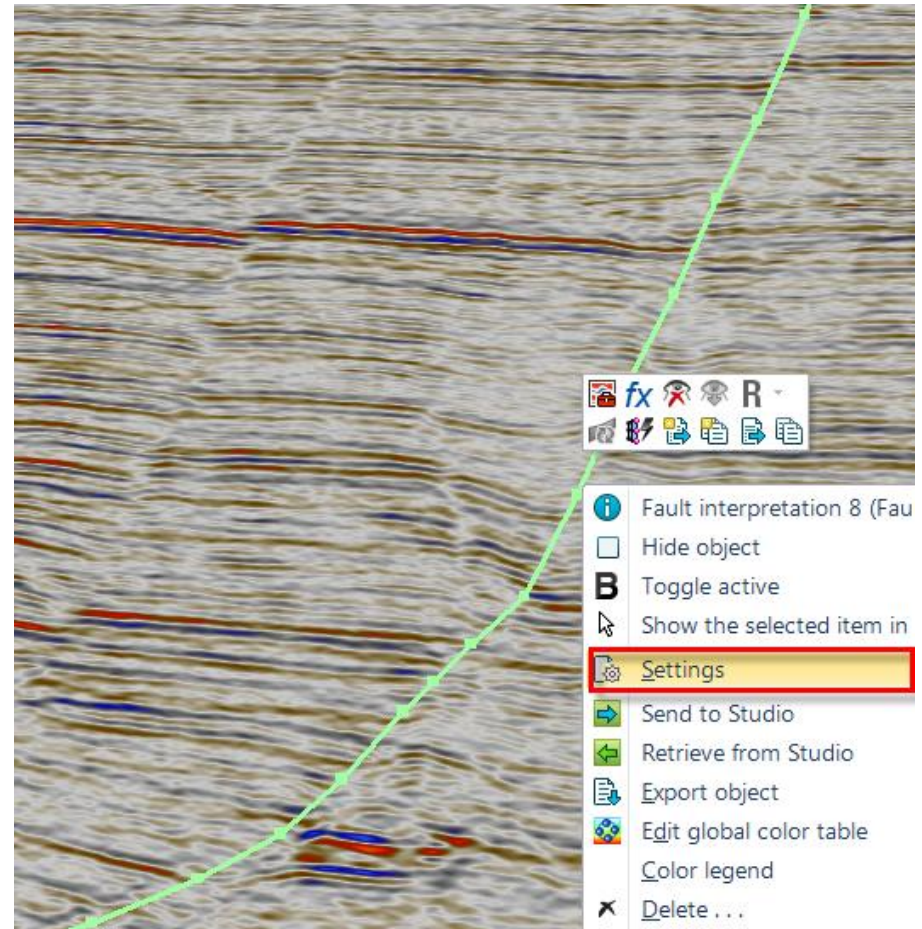
1. Activate a fault in an **Interpretation** window.
2. Right-click this active fault. From the **Tool Palette**, click the *Paint brush* and click the fault segment (changes to yellow).
3. Right-click again from the same fault segment.
4. From the mini toolbar, select how you want the fault to be reassigned.

The new fault appears in the **Input** pane.



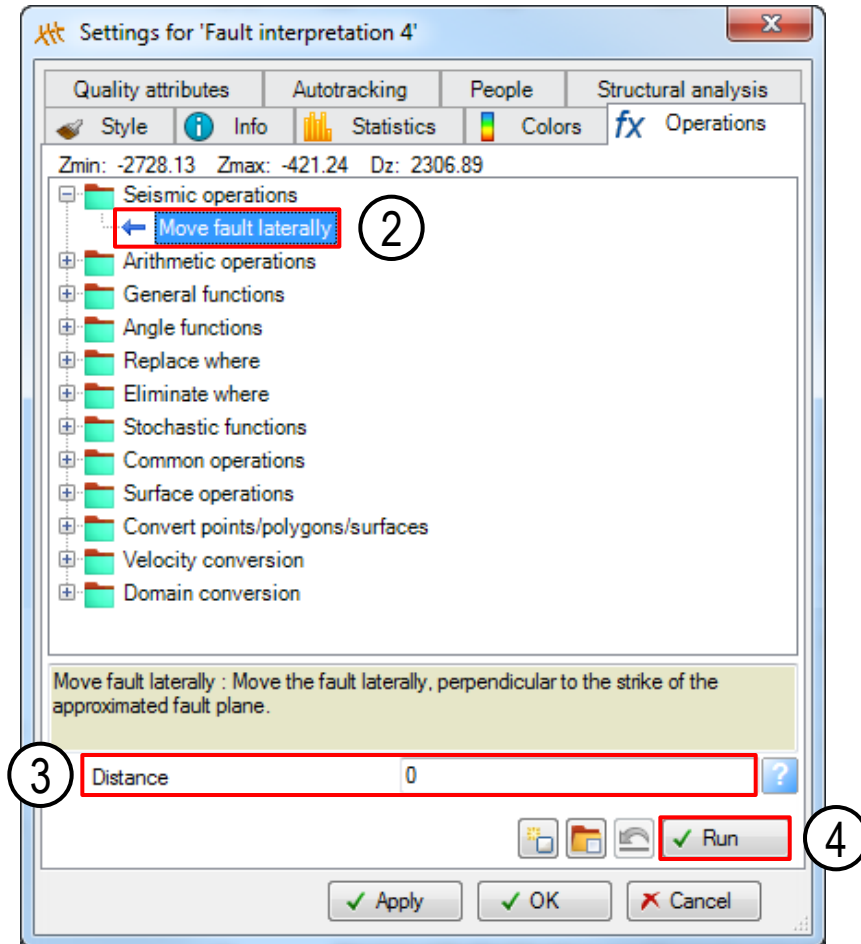
Move faults laterally (1)

1. Right-click the fault and click *Settings*.



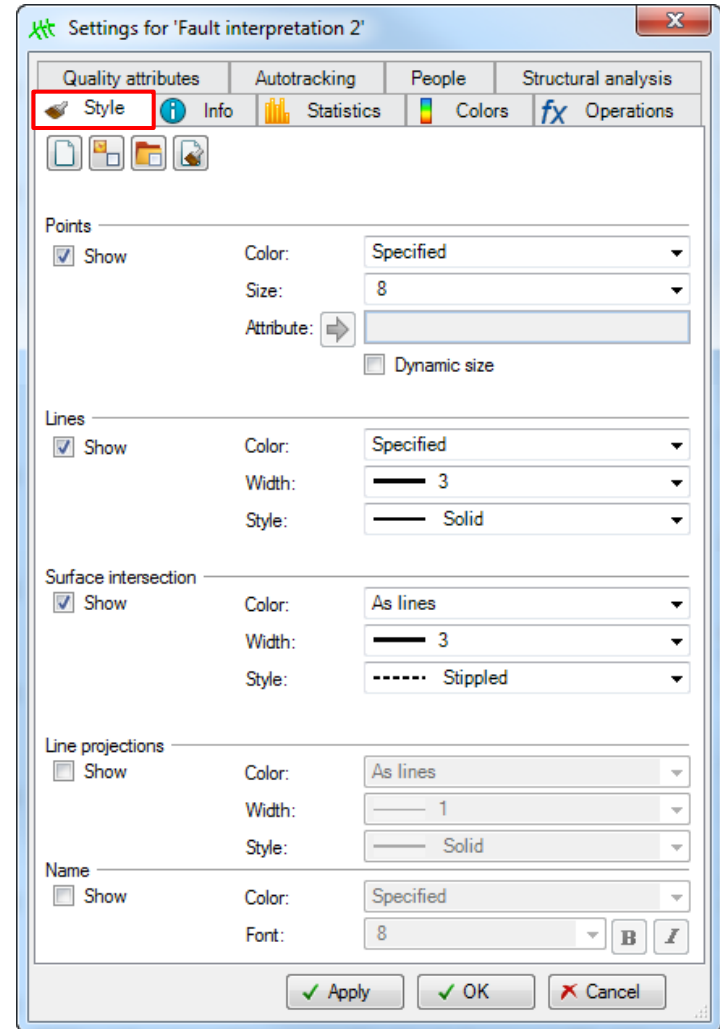
Move faults laterally (2)

2. Browse to the **Operations** tab, under **Seismic operations** subfolder, and select *Move faults laterally*.
3. Set the Distance value.
4. Click *Run*. The active fault is moved laterally west (negative value) or east (positive value).



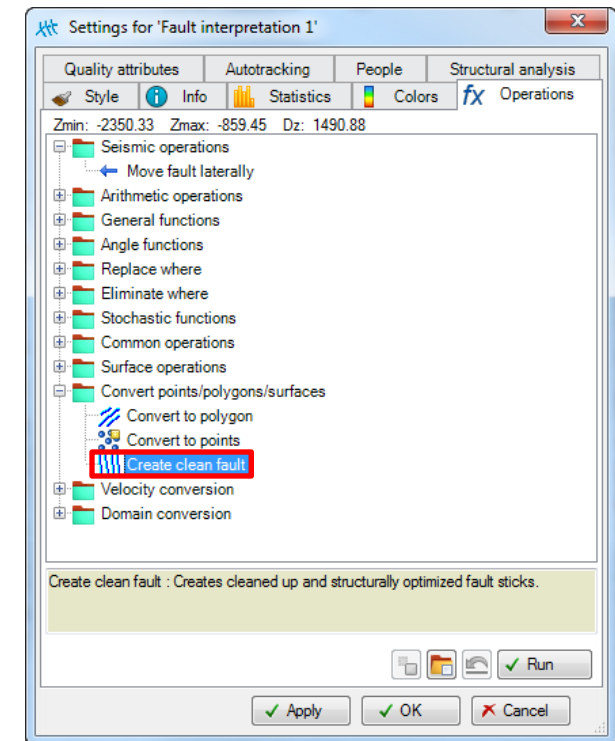
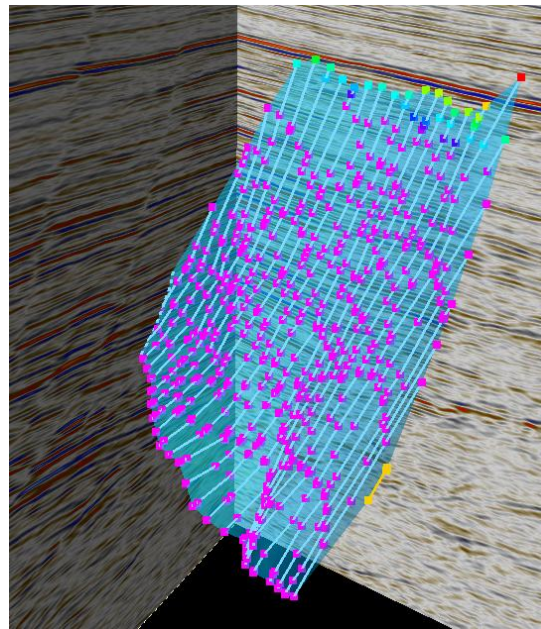
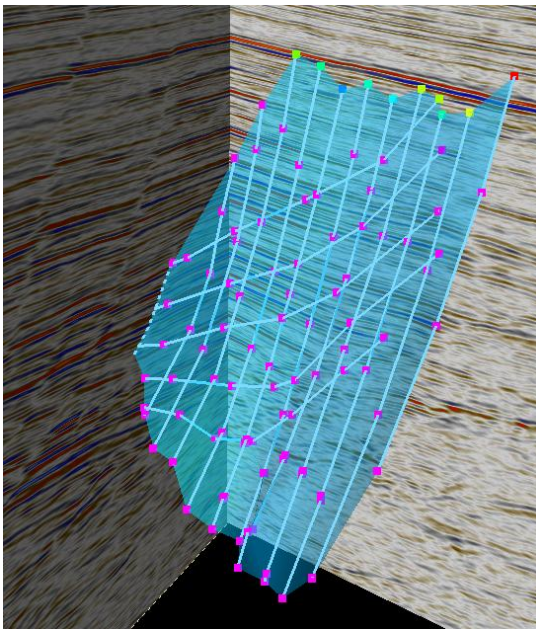
Display and annotate faults

1. Open the **Fault settings** dialog box.
2. On the **Style** tab, change one or more of these options for the fault display settings.



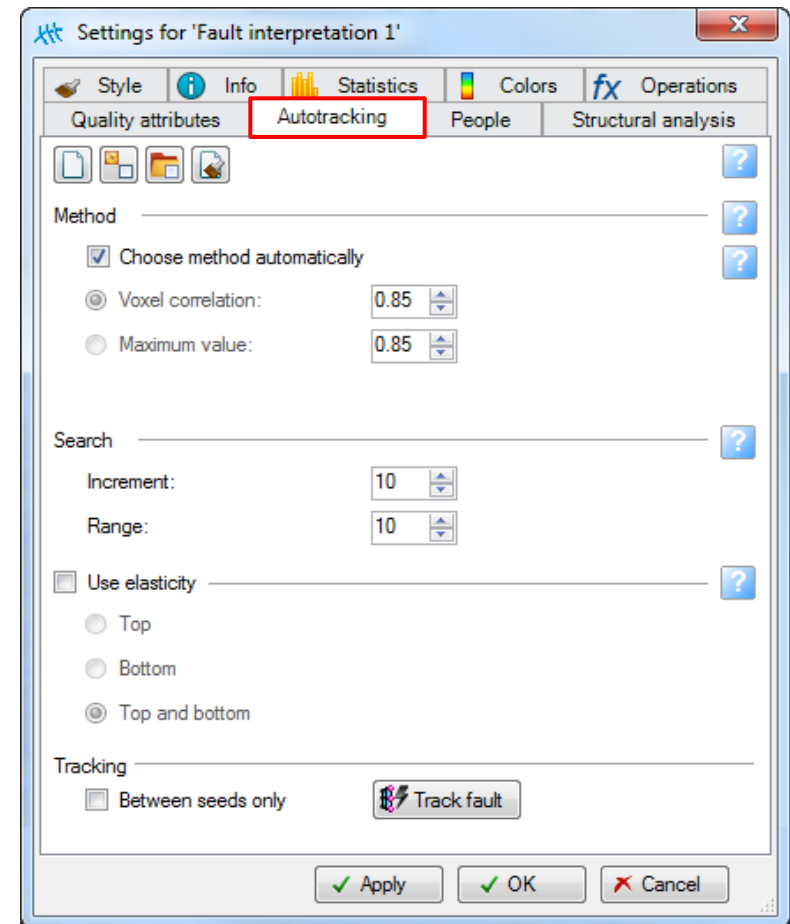
Clean faults

1. On the **Operations** tab in the **Settings for Fault interpretation** window, expand **Convert points/polygons/surfaces**.
2. Select *Create clean fault*.
3. Click *Run*.



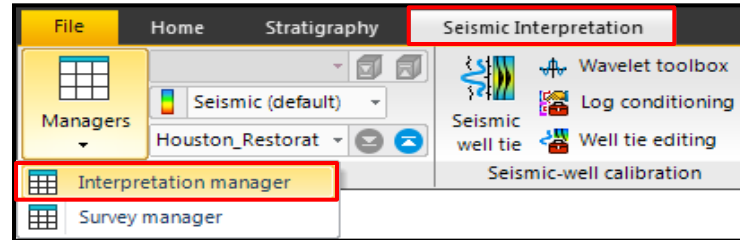
Fault autotracking

If fault geometry is not too complex, Fault autotracking is a useful tool to get a quick view of a fault system or to interpret faults in more detail.

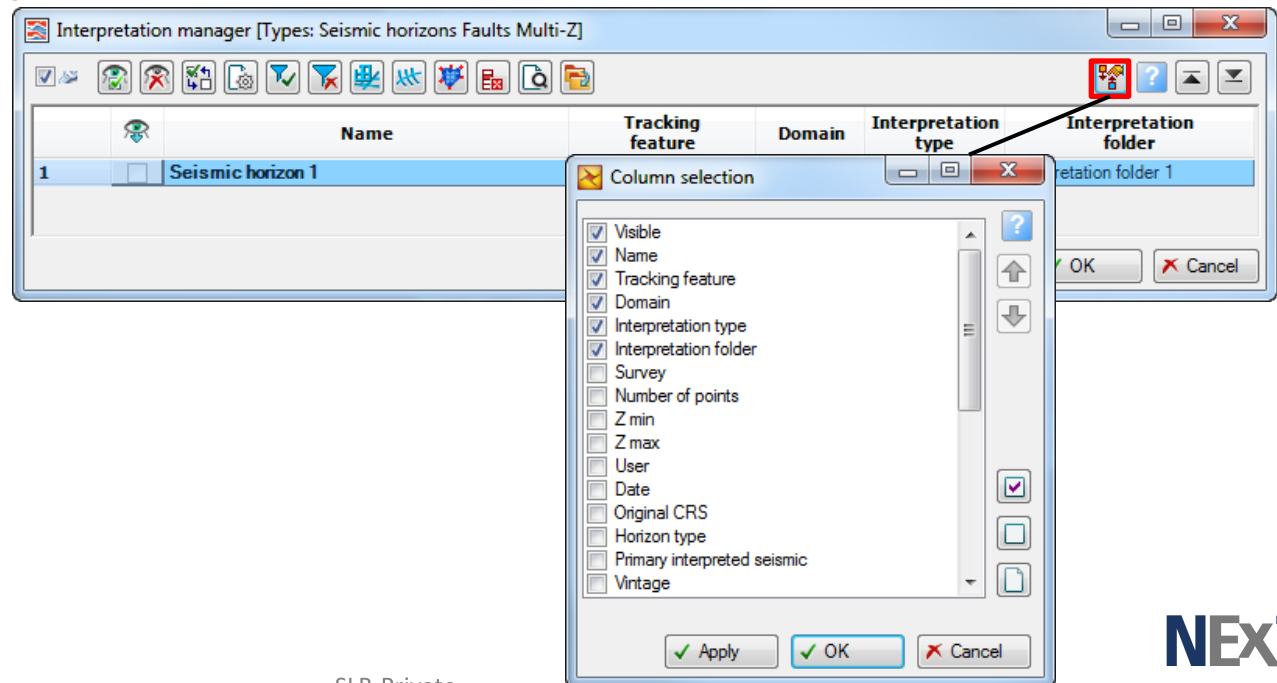


Interpretation Manager

1. On the **Seismic Interpretation** tab, click *Managers* and select *Interpretation manager*.



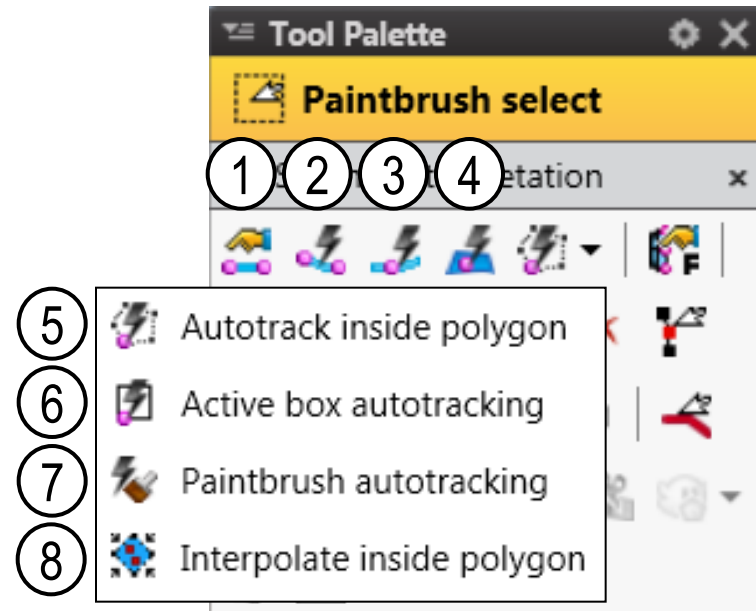
2. In the **Interpretation manager**, click *Select columns to display in Interpretation manager* to control the appearance of the spreadsheet.



Horizon Interpretation Techniques

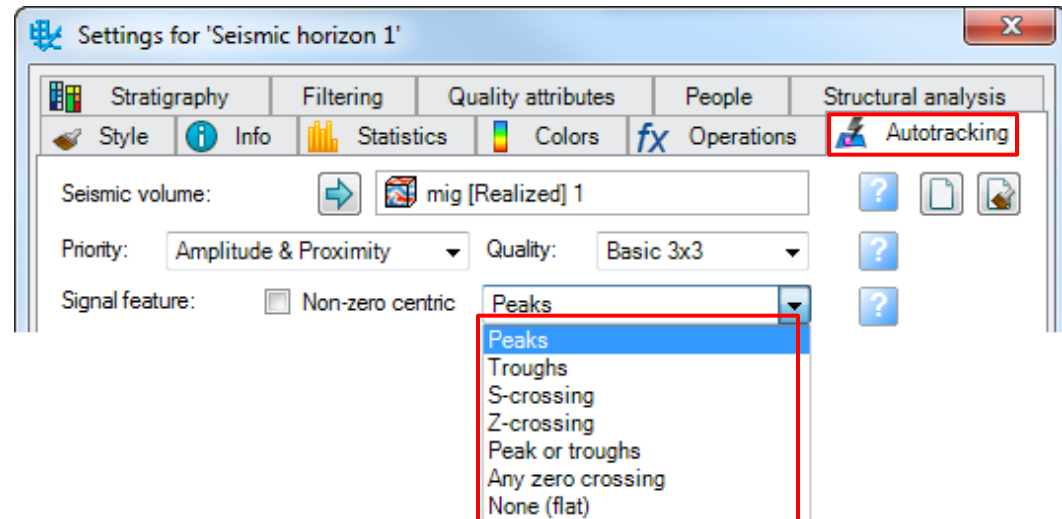
- Horizon interpretation is a combination of many techniques that include manual and automated interpretations.

- 1 Manual Interpretation
- 2 Guided autotracking
- 3 Seeded 2D autotracking
- 4 Seeded 3D autotracking
- 5 Autotrack inside polygon
- 6 Active box autotracking
- 7 Paintbrush autotracking
- 8 Interpolate inside polygon



Manually Interpret a Seismic Horizon (1)

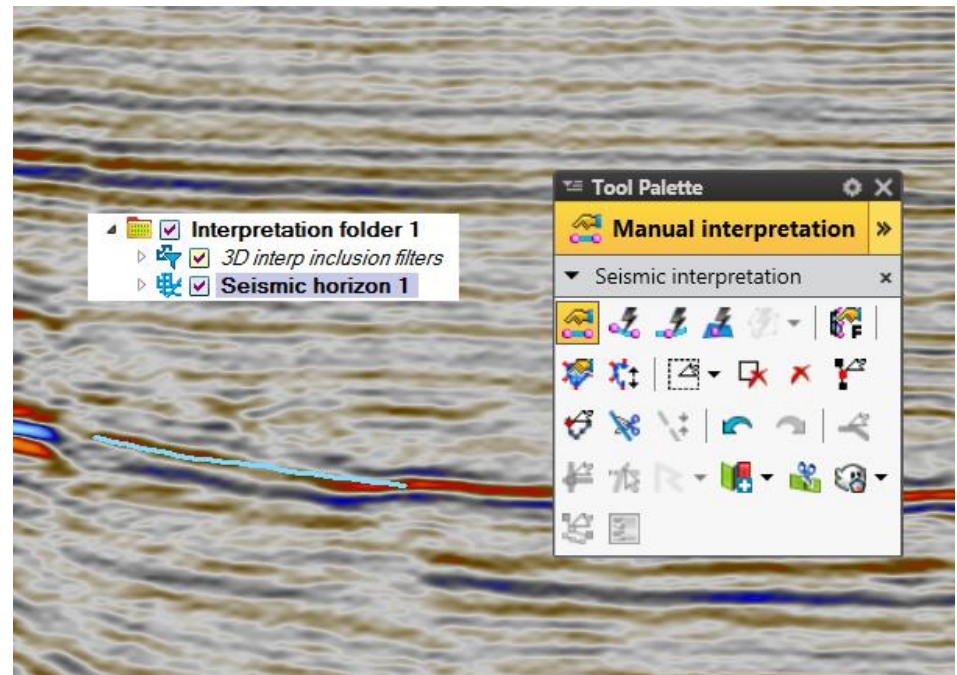
1. Open an **Interpretation** window and display a seismic section.
2. Right-click the seismic section and click *New seismic horizon*  in the mini toolbar.
3. Select the signal feature to track:
 - a. Right-click the seismic horizon in the **Input** pane.
 - b. Open **Settings** and click the **Autotracking** tab.
 - c. Select one of the options from the list.



Note: Your choice works only for 2D and 3D guided tracking.

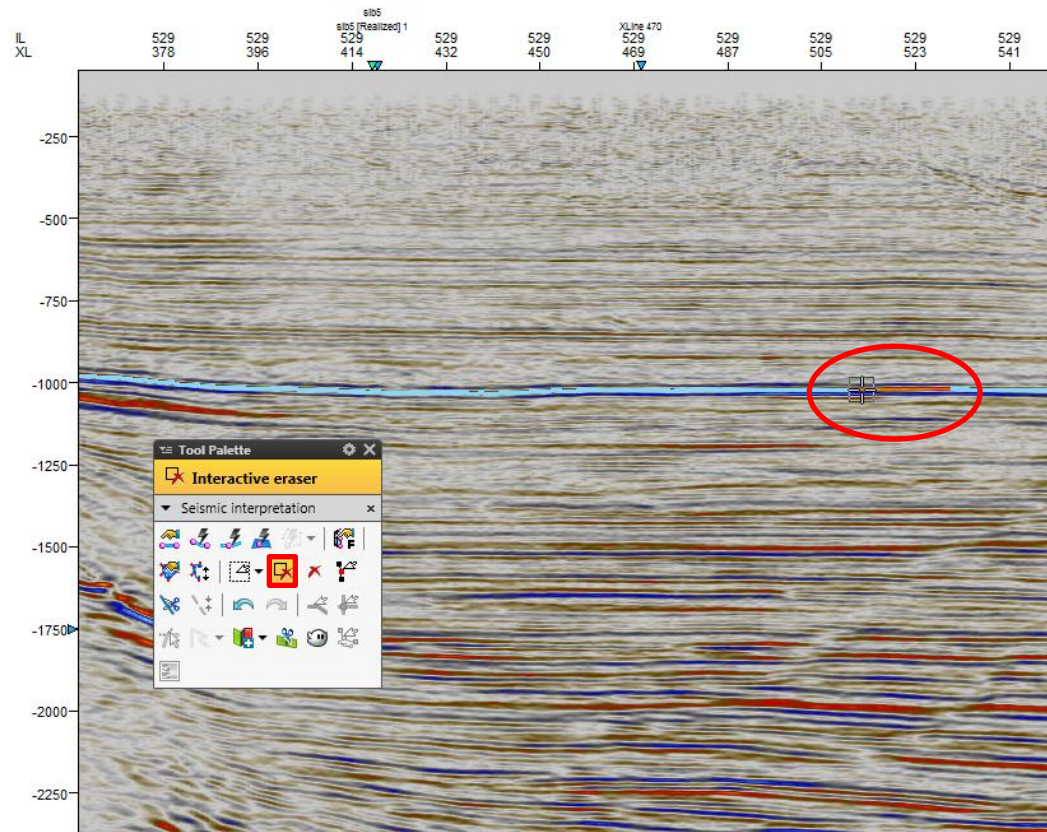
Manually Interpret a Seismic Horizon (2)

4. Click **Seismic Interpretation Tool Palette** in the mini toolbar.
5. Select manual interpretation.
6. Begin to digitize the seismic event on which you want to pick your horizon:
 - a. Click to interpret the first point, move the cursor in the direction of interpretation and click again to make the 1st segment.
 - b. Press N or double-click to break the pick.



Manually Interpret a Seismic Horizon (3)

7. To erase all or part of the horizon on the seismic section, use *Interactive eraser*  in the **Seismic interpretation** tool palette.



Autotracking

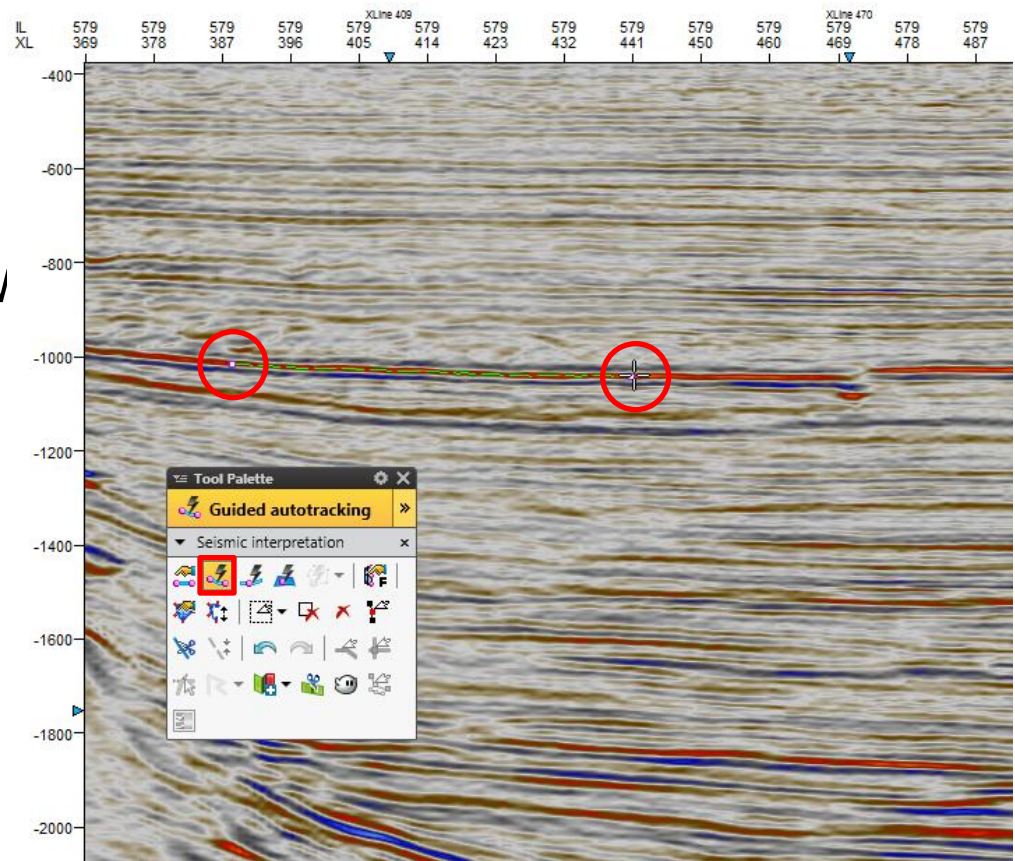
Autotracking is a powerful tool for horizon interpretation, particularly in areas with good reflector continuity, signal strength, and data quality. Choose two or more points; the tracking finds the best route between them.

There are three main types of horizon autotracking:

- Guided Autotracking
- Seeded 2D Autotracking
- Seeded 3D Autotracking

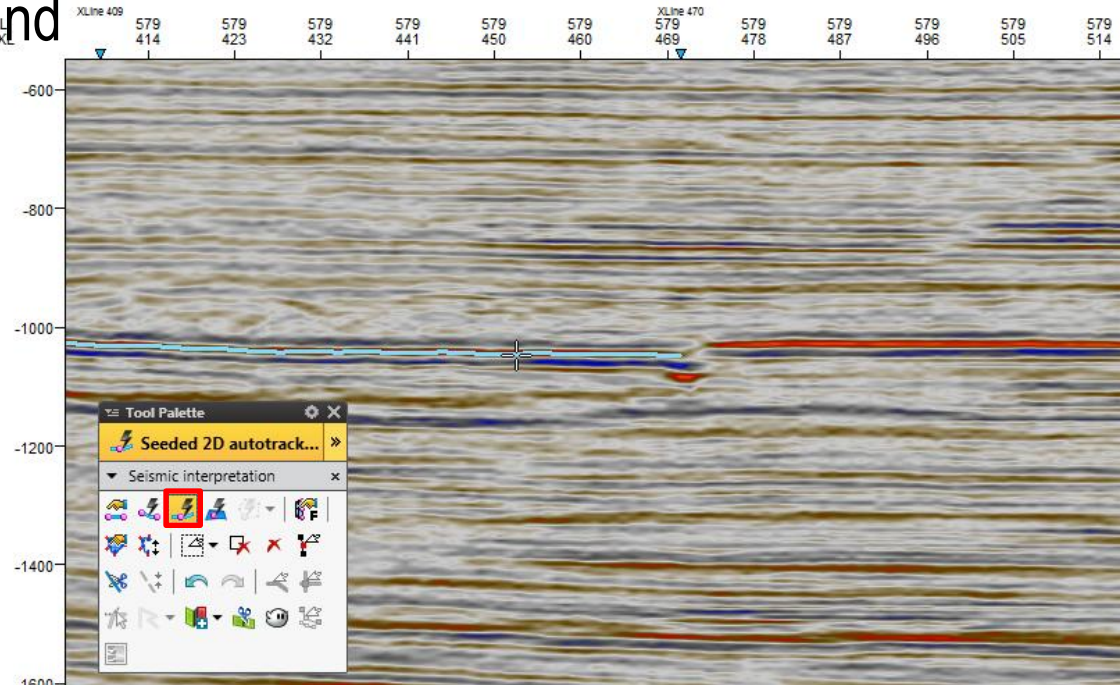
Guided autotracking

1. On the **Seismic interpretation Tool Palette**, click *Guided autotracking*.
2. Click along a reflector to select the start and end points.
 - The interpretation follows the seismic event between the two seed points.
3. Double-click or press N to end an interpretation segment to break the pick.



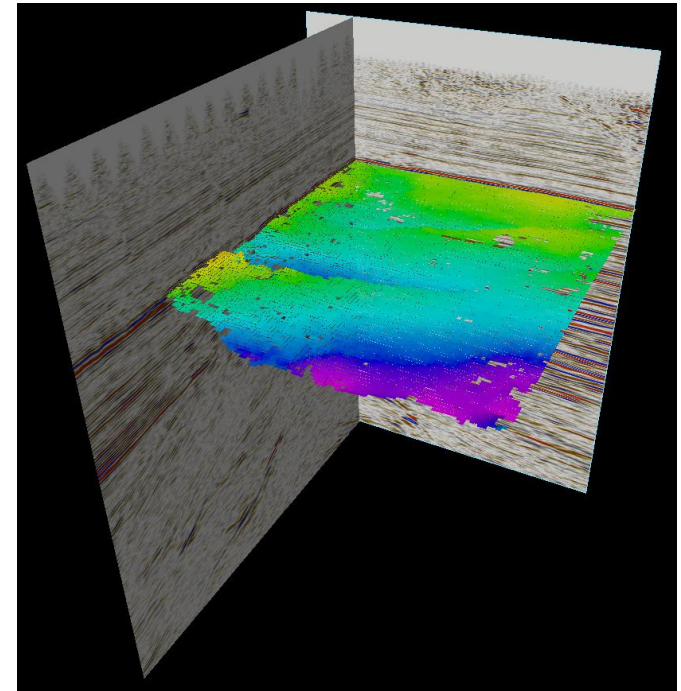
Seeded 2D autotracking

1. Click once along a seismic event. The interpretation follows the event to each side of the selected point and stops when an abrupt change in an amplitude value occurs (or parameters are no longer met).
2. Repeat the procedure for the segment across the fault and other lines.



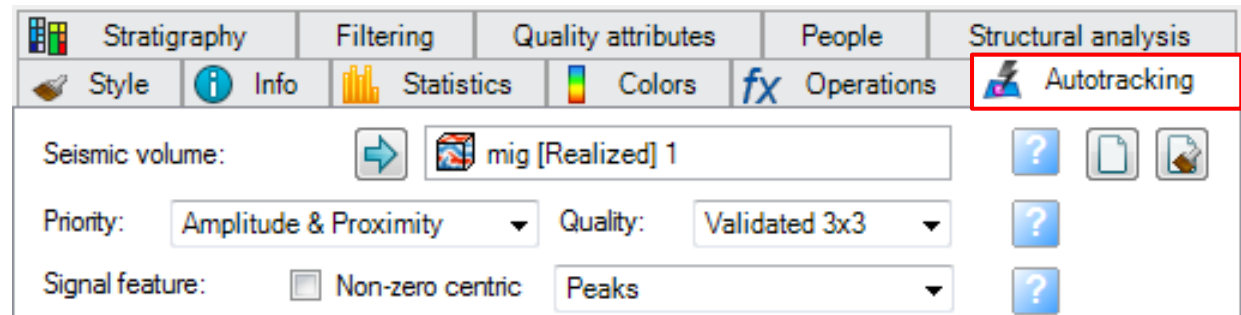
Seeded 3D autotracking

- Works only for 3D seismic data. Real control for Seeded 3D autotracking comes from options on the **Autotracking** tab in the horizon **Settings** dialog box.
- These options allow you to pick horizons that require minimum quality control from the interpreter.
- Choose one or several seed points and the horizon is tracked outward in all directions from these seed points.
- When the reflectors are of good quality, this method is an efficient way of interpreting through the seismic cube.



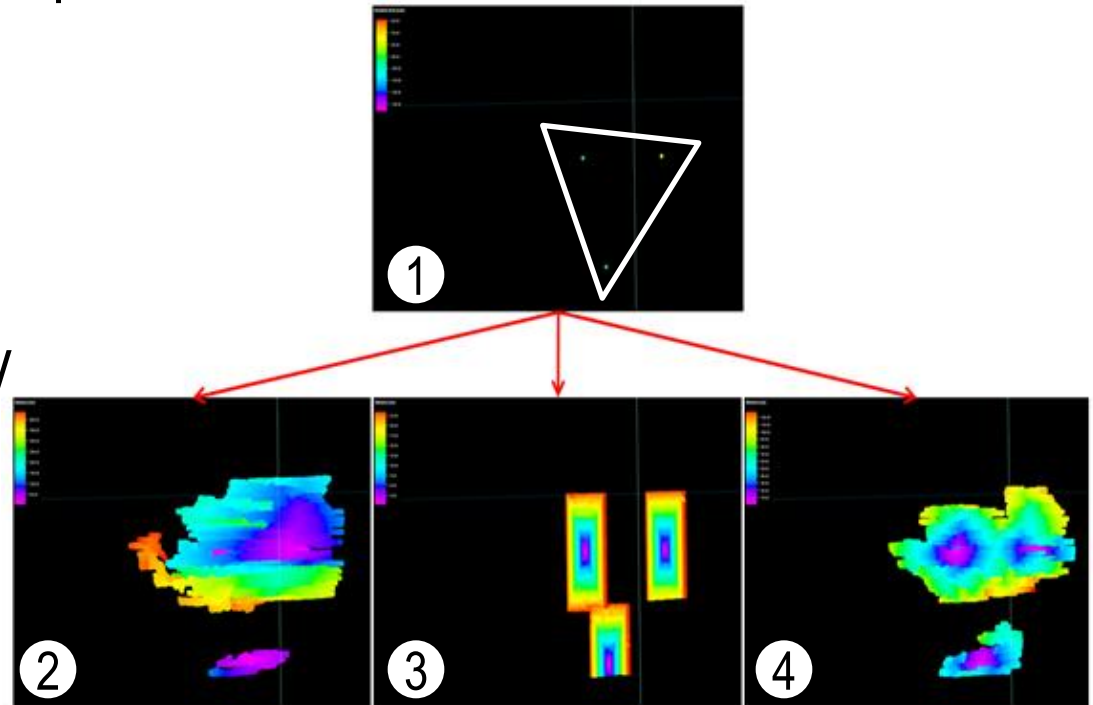
Autotracking tab: General section

- The General area on the **Autotracking** tab allows you to select:
- Seismic volume (picked automatically)
- Priority
- Quality
- Seismic volume



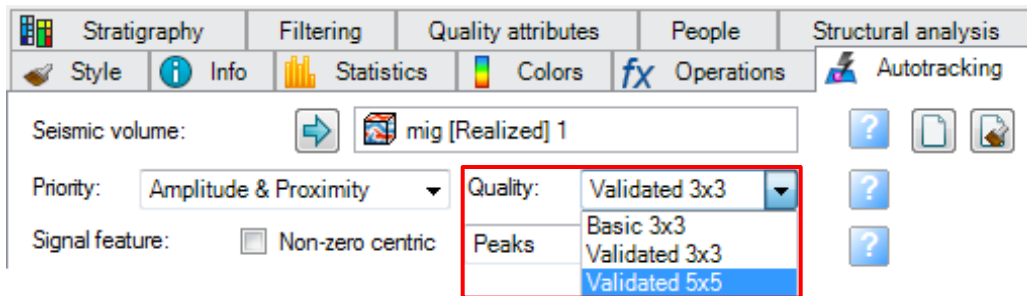
Autotracking tab: Priority

- The Priority option regulates the priority queue of available seed points for the expansion process.
- The available options in the process:
 - 1 Original seeds
 - 2 Amplitude
 - 3 Proximity
 - 4 Amplitude and proximity



Autotracking tab: Quality

- The Quality option on the **Autotracking** tab controls how stringently each seed point is checked before it is expanded.



1 Basic 3x3

2 Validated 3x3

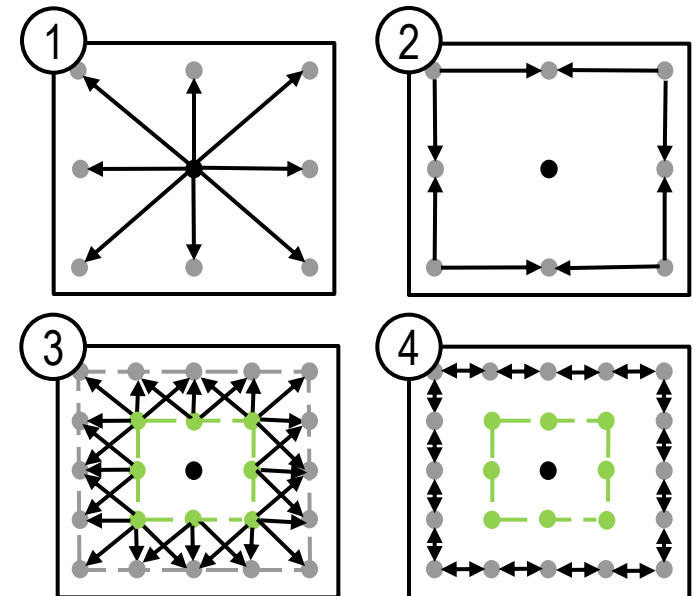
3 Validated 5x5 (3rd stage)

4 Validated 5x5 (4th stage)

● Seed point

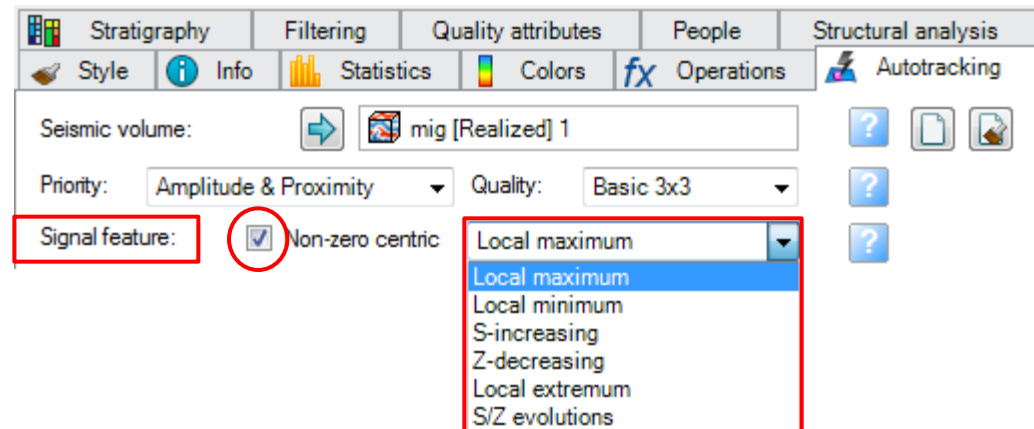
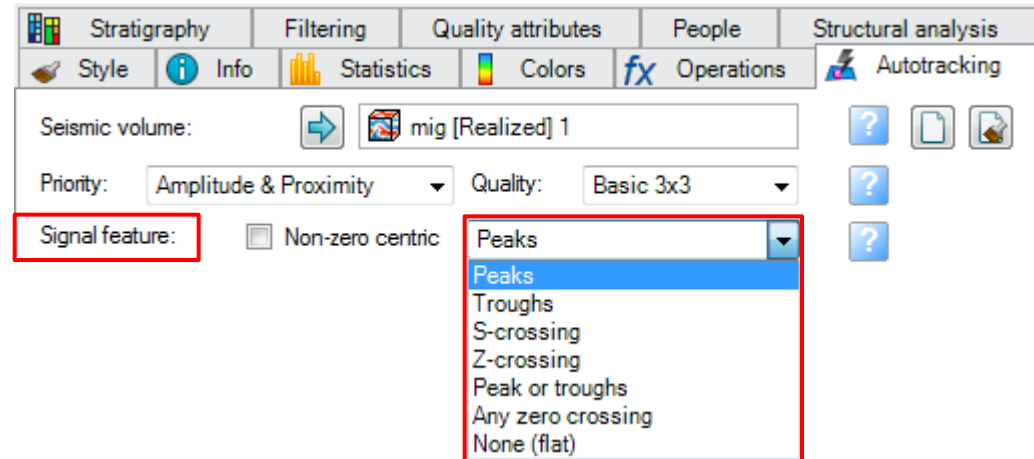
● Potential child points

● Accepted child points from phases 1 and 2



Autotracking tab: Signal feature (1)

- *Signal feature* controls the feature to be tracked.



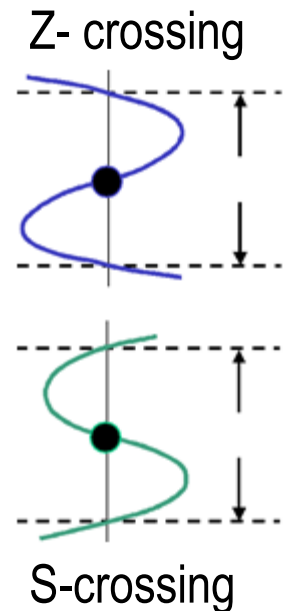
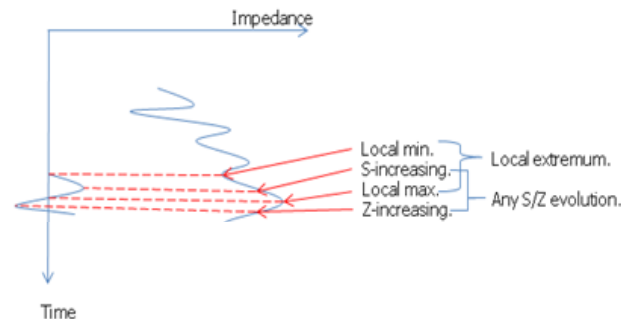
Autotracking tab: Signal feature (2)

The illustration shows what *non-zero centric* means.

A time-based SEG-Y plots along the time axis.

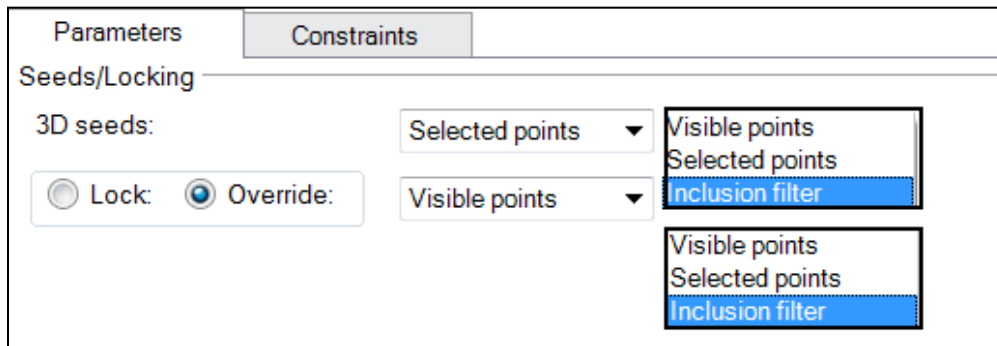
A zero crossing for this represents a local minimum or local maximum in the non-zero centric world.

A peak in the zero-centric seismic represents an S-increasing feature, while a trough represents a Z-increasing feature.



Autotracking tab: Seed/Locking

- Use visible or selected points as 3D seeds. Choose to lock or override the existing interpretation.

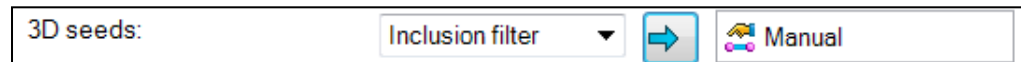


Parameters Constraints

Seeds/Locking

3D seeds: Selected points Visible points
Selected points
Inclusion filter

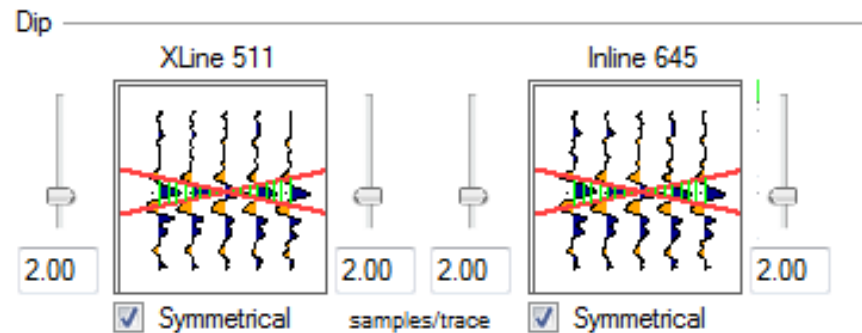
☐ Lock: ☒ Override: Visible points Visible points
Selected points
Inclusion filter



3D seeds: Inclusion filter → Manual

Autotracking tab: Parameters subtab - Dip/Wavelet

Dip: Optimize the autotracking parameters for dipping reflectors.

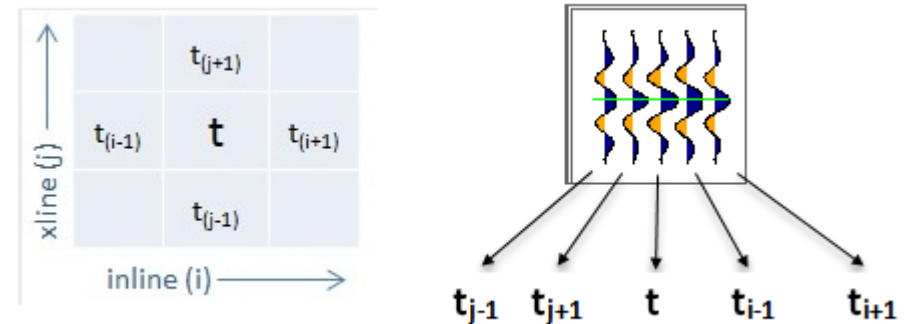
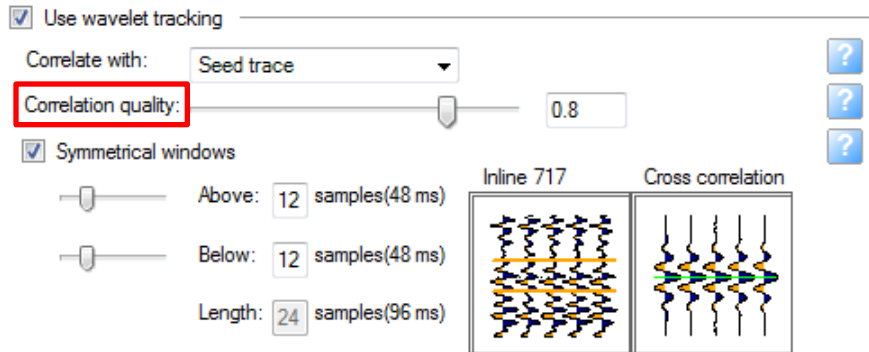


Wavelet: Test the potential expansion points using the correlation of a seismic trace window.

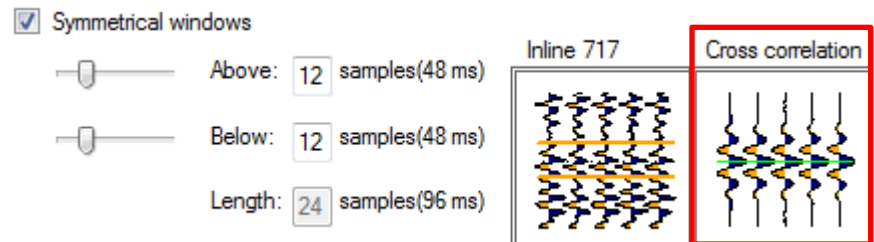
☒ Use wavelet tracking

Correlate with: Seed trace

Autotracking tab: Wavelet



Cross correlation live diagram



Autotracking tab: Parameters subtab - Doublet/Tracking rule

Specifies which part of a doublet the tracker follows.

Doublet

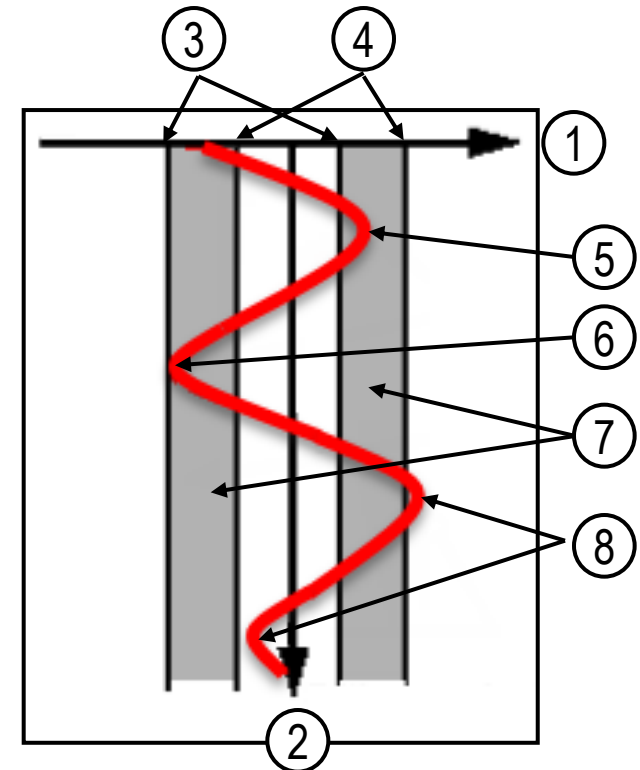
☒ Tracking rule:

Lower

Lower

Upper

- | | | | |
|----------|-------------------------------|----------|------------------------|
| 1 | Amplitude/reflection strength | 5 | Accepted maximum event |
| 2 | Z (time or depth) | 6 | Accepted minimum event |
| 3 | Minimum attribute | 7 | Legal event region |
| 4 | Maximum attribute | 8 | Rejected event |



Autotracking tab: Constraints subtab (1)

Lateral limits: Limit the autotracking horizontally.

Vertical limits: Constrain the autotracking vertically.

The screenshot shows the 'Constraints' subtab of the Autotracking interface. It is divided into two main sections: 'Lateral limits' and 'Vertical limits'. The 'Lateral limits' section includes a 'Boundary polygon' checkbox and a 'Stop at visible faults' checkbox. The 'Vertical limits' section includes checkboxes for 'Upper', 'Lower', 'Vertical range', and 'Trend data', each with a corresponding input field and a help icon. The 'Upper' and 'Lower' options are set to 'Upper horizon' and 'Lower horizon' respectively, with offsets of 100 ms and 10 ms. The 'Vertical range' is set to -1100 to -950. The 'Trend data' is set to 'Seeded 3D Autotracking Horizon'.

Parameters Constraints

Lateral limits

☒ Boundary polygon: Boundary Polygon

☒ Stop at visible faults

Vertical limits

☒ Upper: Upper horizon Offset: 100 ms ?

☒ Lower: Lower horizon Offset: 10 ms ?

☒ Vertical range: -1100 -950 ?

☒ Trend data: Seeded 3D Autotracking Horizon ?

Using co-volume: If selected, the autotracking uses two seismic inputs instead of one.

The screenshot shows the 'Use co-volume' section of the Autotracking Constraints subtab. It includes a 'Use co-volume' checkbox, a 'Signal feature' dropdown menu, a 'Seismic co-volume' input field, a 'Usage' dropdown menu, and a 'Value range' input field. The 'Use co-volume' checkbox is checked. The 'Signal feature' is set to 'Non-zero centric' with a 'Peaks' dropdown. The 'Seismic co-volume' is set to 'mig [Amplitude contrast] [Realized] 1'. The 'Usage' is set to 'Track if point exists on both volumes'. The 'Value range' is empty.

☒ Use co-volume

Signal feature: ☐ Non-zero centric Peaks ?

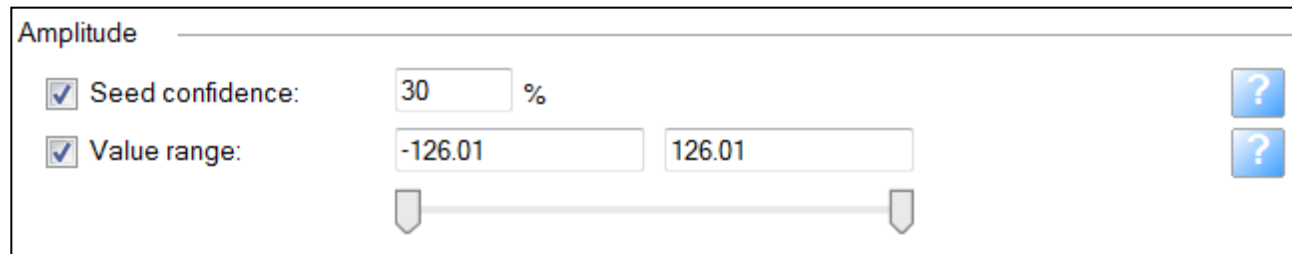
Seismic co-volume: mig [Amplitude contrast] [Realized] 1 ?

Usage: Track if point exists on both volumes ?

☐ Value range: ?

Autotracking tab: Constraints subtab (2)

Seed confidence: Specifies the minimum value for the seismic amplitude as a percentage of the seed point.



The screenshot shows a software interface for the 'Amplitude' section. It contains two checked options: 'Seed confidence' and 'Value range'. The 'Seed confidence' option has a text input field with the value '30' and a '%' symbol. The 'Value range' option has two text input fields with values '-126.01' and '126.01'. Below these fields is a horizontal slider bar with arrows at both ends. To the right of each option is a blue square button with a white question mark.

Value range: Allows you to specify a minimum and maximum amplitude value seismic attribute to be tracked. Picks outside these amplitude limits are rejected.

Paintbrush autotracking

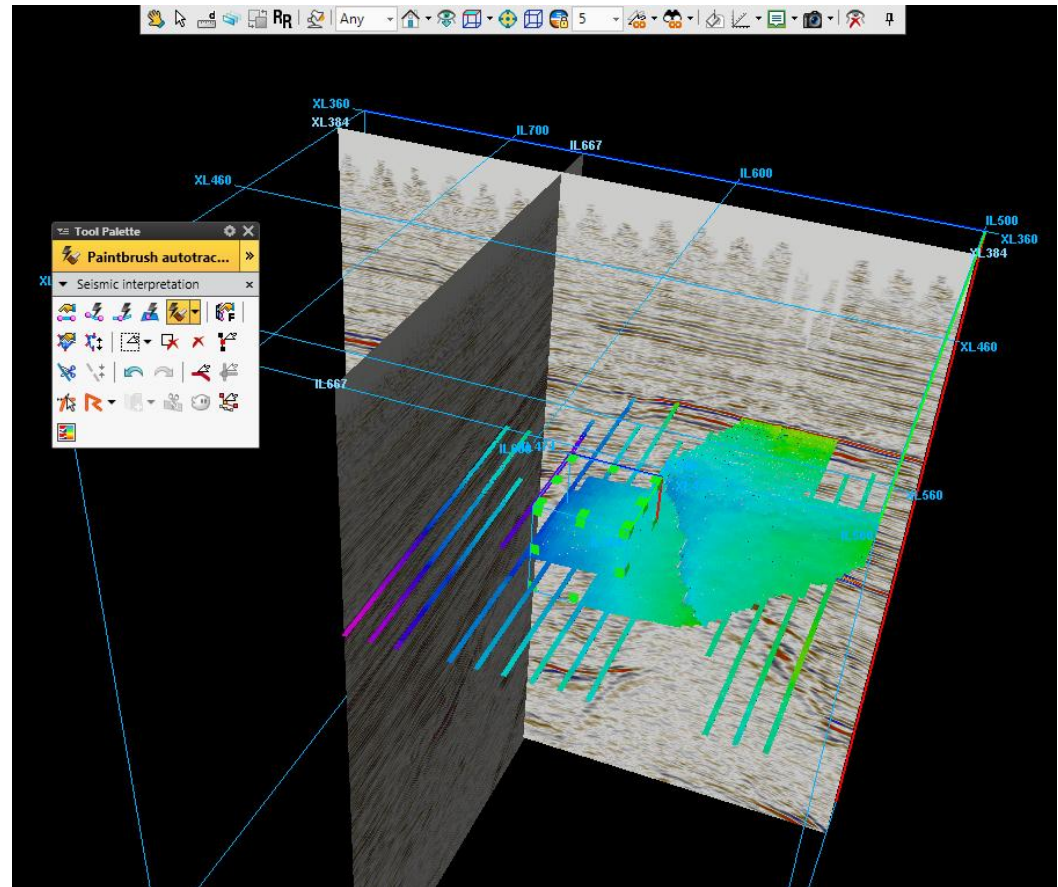
- In Paintbrush autotracking, points are tracked outward from the seed points, limited by the active box size.
- Drag the cursor to track a horizon interpretation that follows the cursor path.
- This tool works in both 2D and 3D windows in slightly different ways.

Perform Paintbrush autotracking in a 3D window (1)

1. Open a new **3D window**.
2. Display an active horizon interpretation in the **3D window**.
(Optional: Make a partial interpretation in an Interpretation or **3D window**.)
3. Right-click the seismic volume used as the source of the autotrack and click *Insert virtual cropped volume* from the menu.
4. Downsize the cropped volume around some of the interpretation.
5. On the **Seismic interpretation Tool Palette**, click *Paintbrush autotracking*.

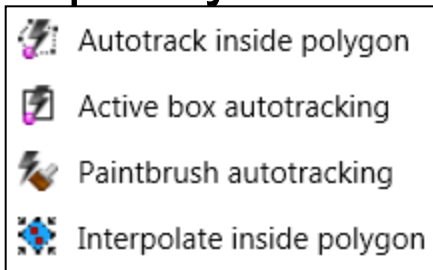
Perform Paintbrush autotracking in a 3D window (2)

6. Drag the mouse on the (transparent) side of the cropped volume. (Make sure the seed points are captured by the first selection inside the cursor volume.)



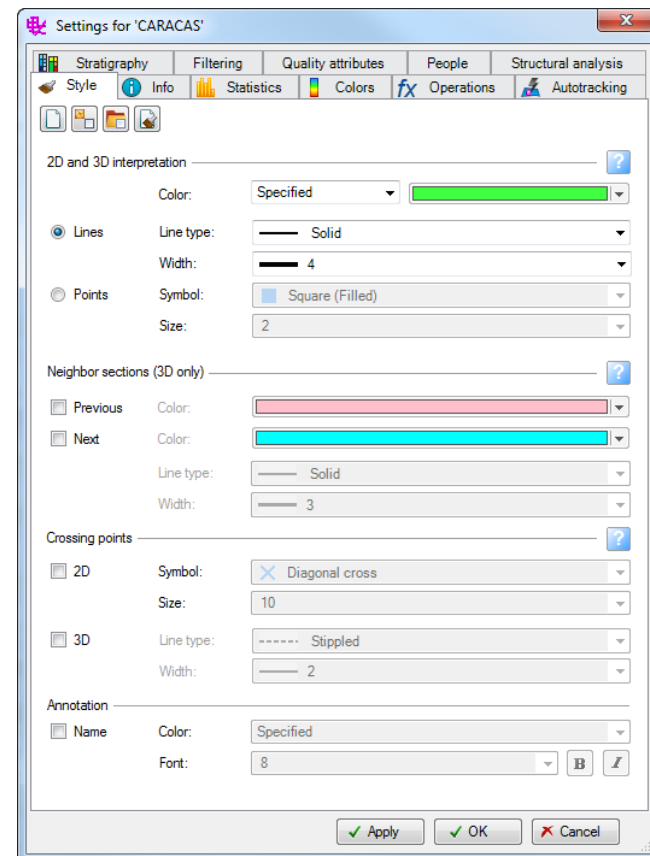
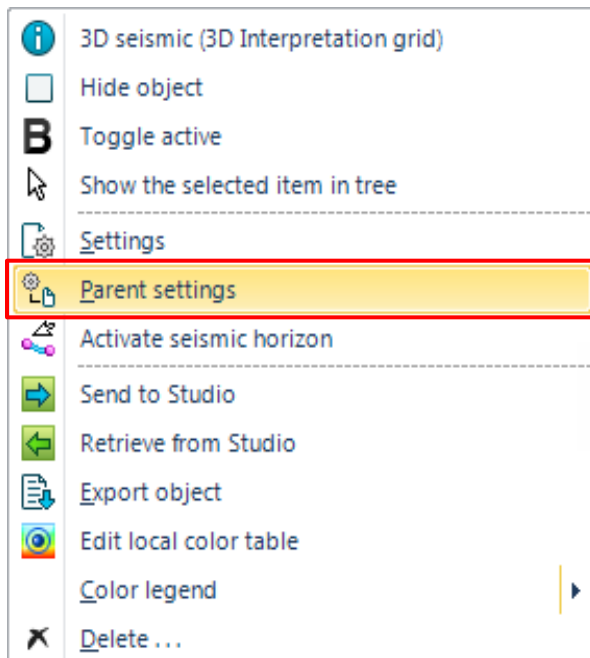
Active box autotracking/Autotrack and Interpolate inside polygon

- *Active box*: Design a box and do autotracking inside, only in a **2D window**.
- *Autotrack inside polygon*: Draw a polygon and do autotracking inside.
- *Interpolate inside polygon*: Remaining gaps in a 3D horizon interpretation can be quickly filled in within a digitized polygon.



Display and annotate a horizon in an Interpretation window

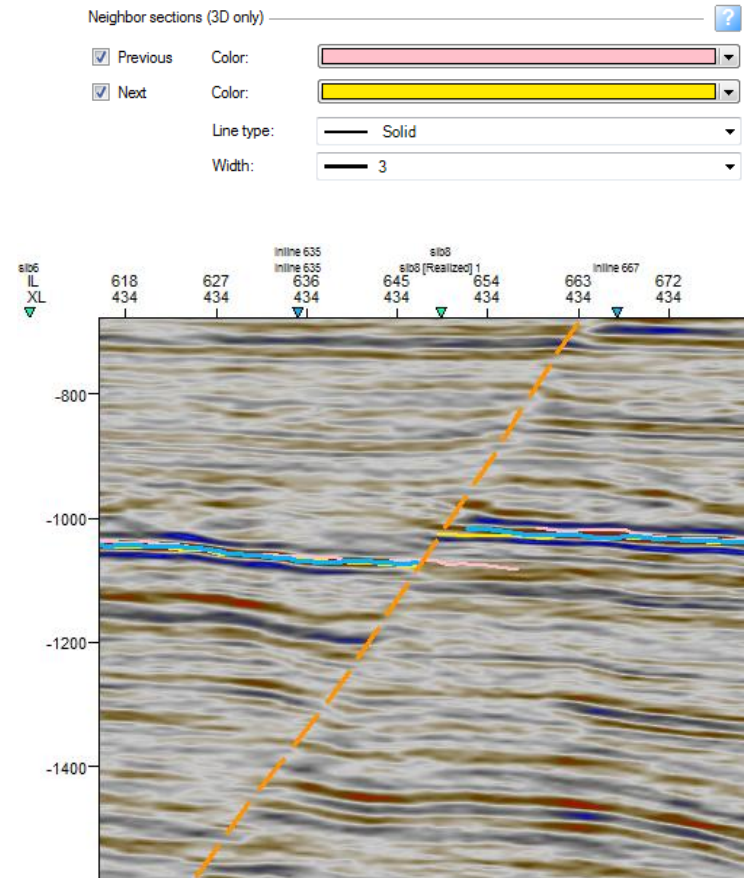
1. Display the horizon interpretation in the **Interpretation** window.
2. Right-click the horizon and click *Parent settings*.
3. Set the display style for 2D/3D interpretation and crossing points.



Display and annotate horizon: Interpretation window (2)

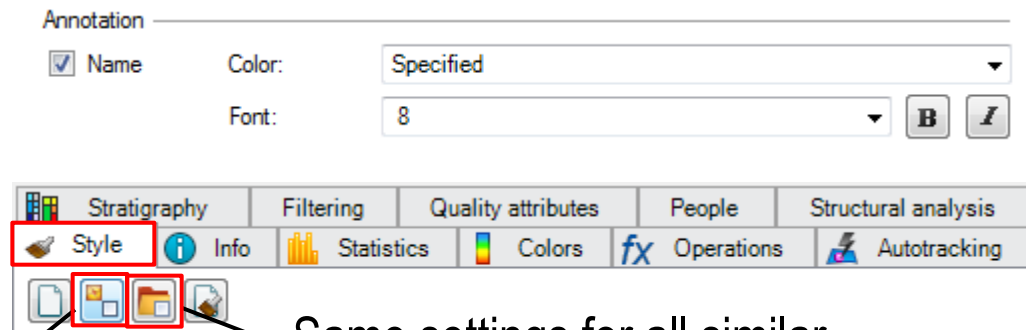
- To aid during interpretation and to know where you picked the horizon on the previous or next intersection, click *Neighbor sections*.

- Note:** This option is available only for 3D data.



Display and annotate horizon: Interpretation window (3)

5. In the **Annotation** section, choose to display the name of the horizon on the section.
6. To apply the same settings for all the Horizons in your project, click one of two options on the **Style** tab.

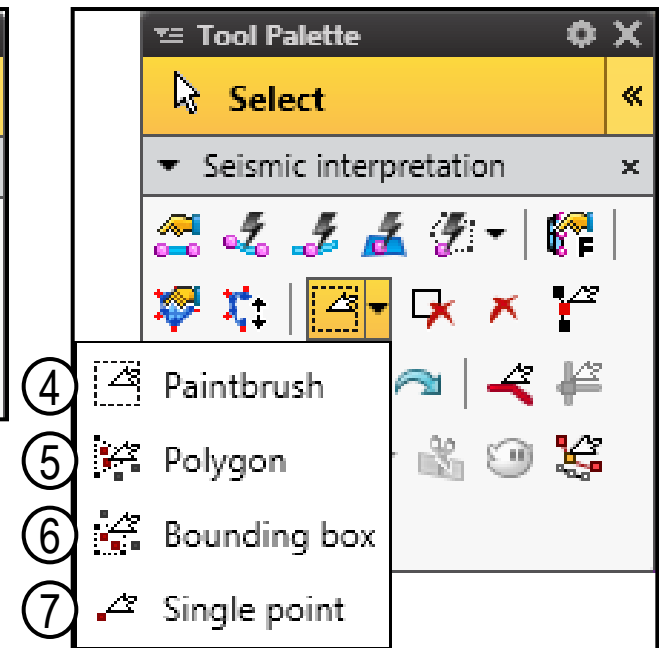
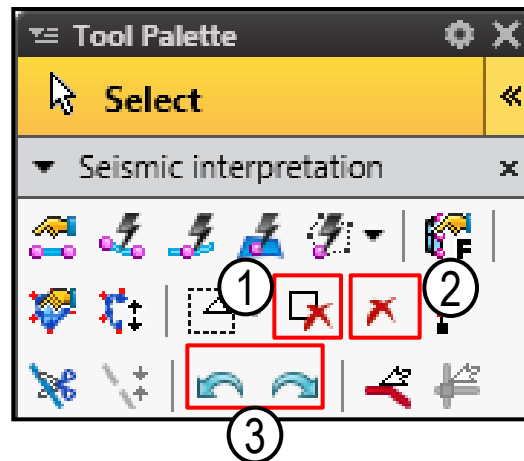


Same settings for all similar objects in project

Same settings for all similar objects in same folder as object

Edit and manipulate using the Seismic Interpretation Tool Palette

- 1 Undo (Ctrl+Z)/Redo (Ctrl+Y)
- 2 Interactive eraser
- 3 Delete
- 4 Paintbrush
- 5 Polygon
- 6 Bounding box
- 7 Single point

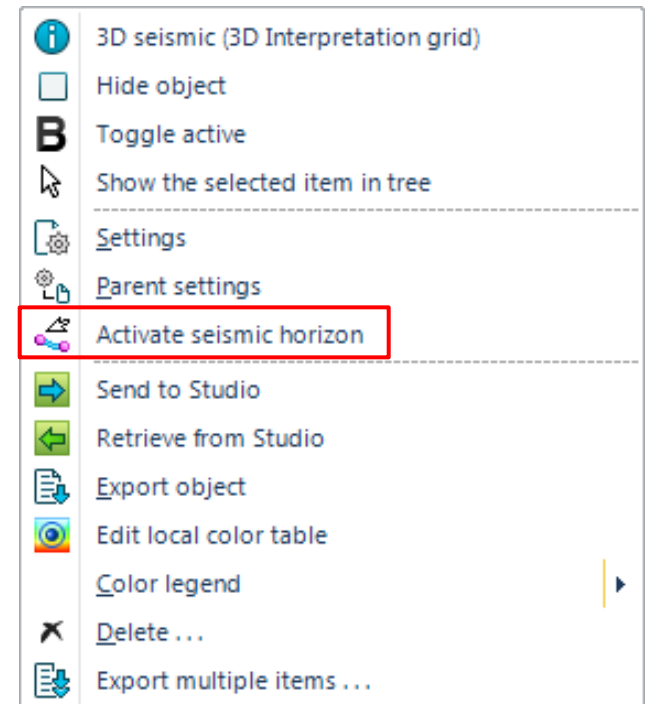


Parent-child editing

- The autotracking uses a trace-to-trace method to verify and expand the seed point.
- When a point or a selection of points is accepted, it becomes a parent to other potential child points that are picked from neighboring traces.

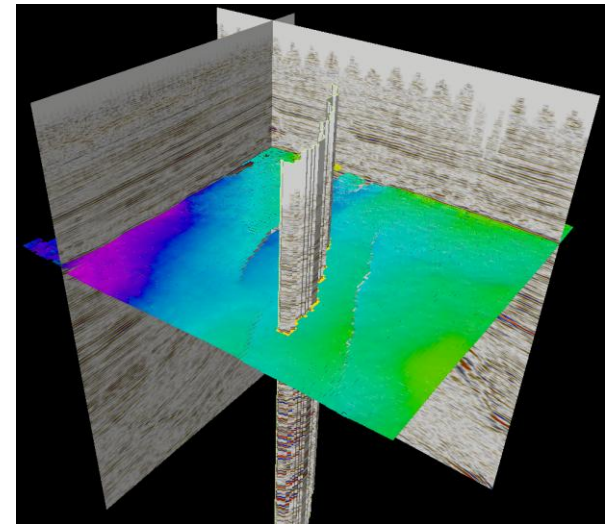
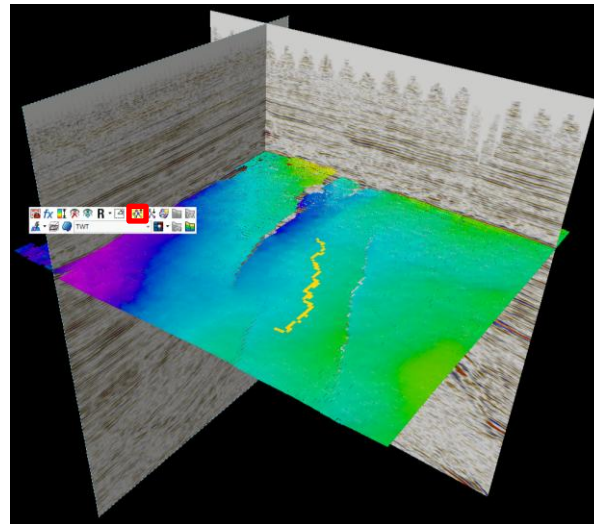
Edit parent and child points in an interpretation (1)

1. Open a **3D window** and display an interpreted horizon from the **Input** pane.
2. Right-click the interpreted horizon and click *Activate seismic horizon*.



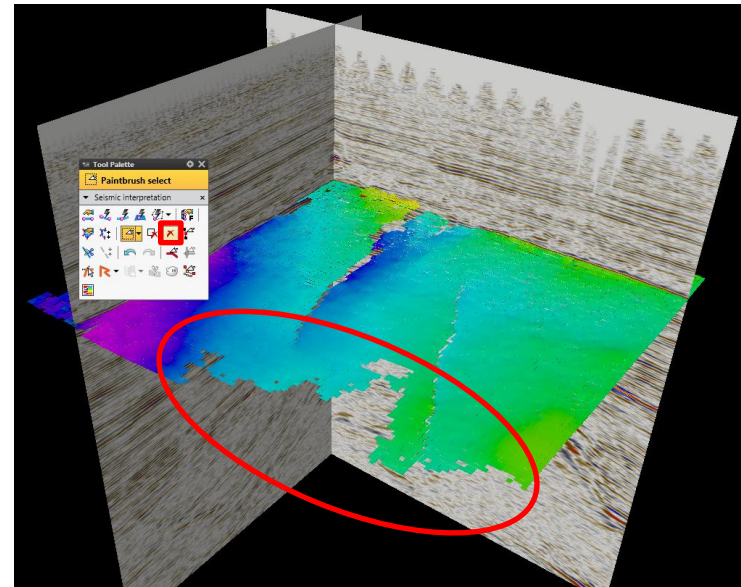
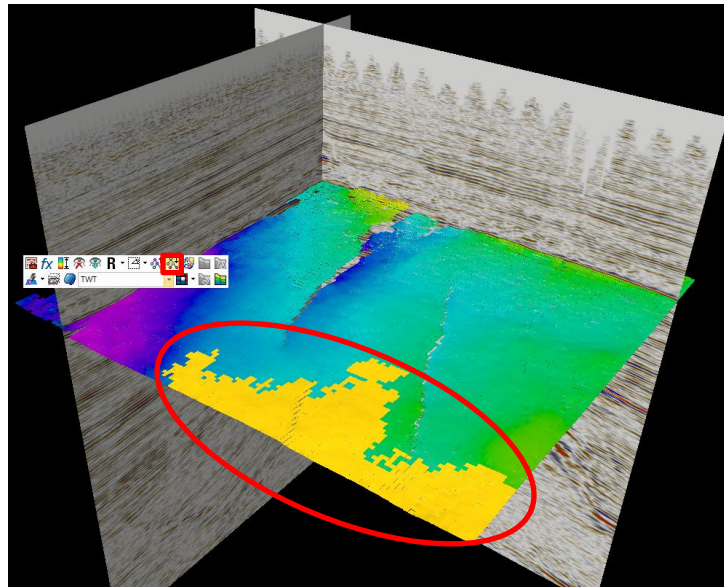
Edit parent and child points in an interpretation (2)

3. Click *Select parent points* in the mini toolbar. When you click a misinterpreted point, the path back to its original seed point turns yellow.
4. To create a fence with the Parent path, press the number 1 key while pointing at the path. You can display this fence in an **Interpretation** window to aid the QC process.



Edit parent and child points in an interpretation (3)

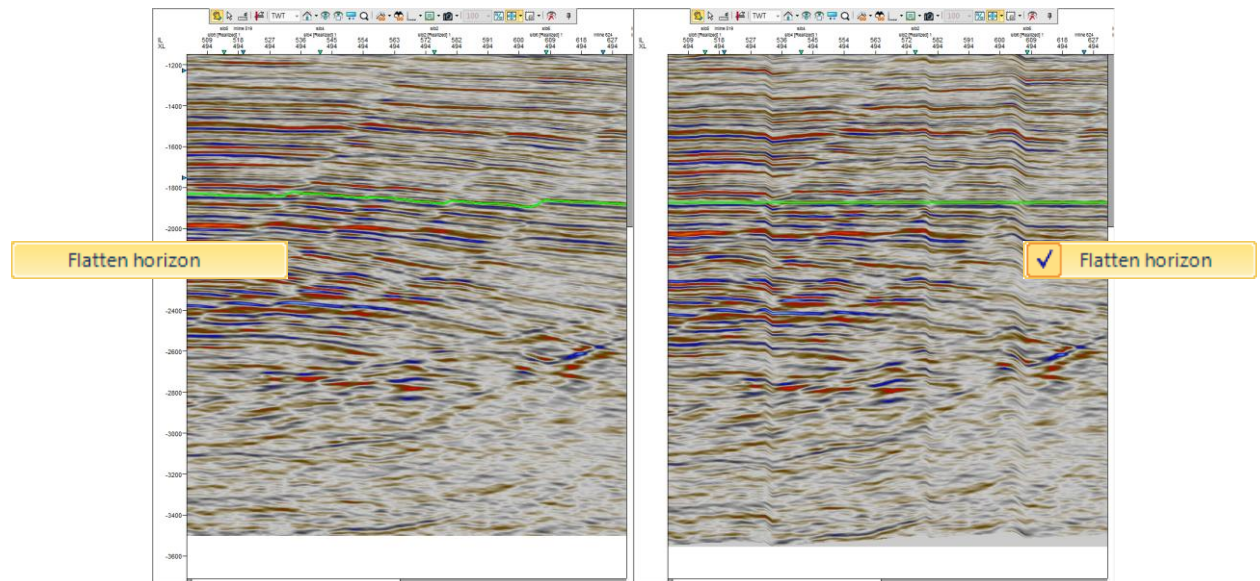
5. Click *Select child point* in the mini toolbar. When you click the first wrongly tracked point along the path, all points originating from it change to yellow.
6. Click *Delete* on the **Tool Palette**. The yellow points are deleted.



Flatten a seismic horizon in Interpretation window

1. Display the horizon in the **Interpretation** window.
2. Right-click and select *Flatten horizon*.
3. To unflatten the horizon, right-click the horizon and switch off *Flatten horizon*.

Note: Petrel allows for gaps in the interpretation, such as breaks over faults.

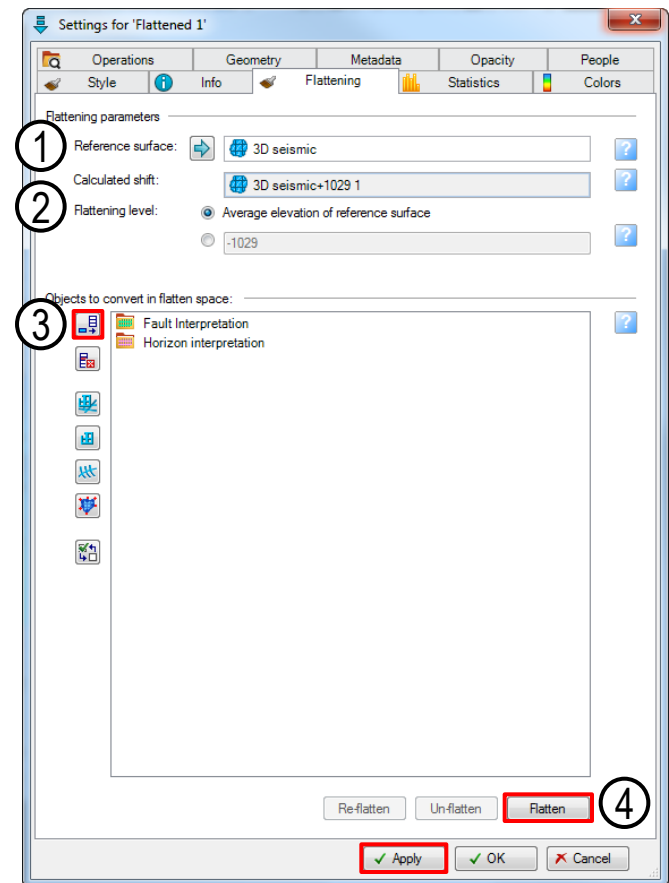


Flattened objects

- Interpretation objects can be converted in the flattened space. These objects can be faults, horizons, surfaces, points, and grid interpretations. (Interpretations are sub-objects of the horizon interpretation, and they are associated with only one seismic survey.)
- Create new interpretation objects (faults, horizons, and grid interpretations) and interpret them directly in the flattened space.
- A history of the flattened objects is stored on the **History** tab in the **Settings** dialog box.

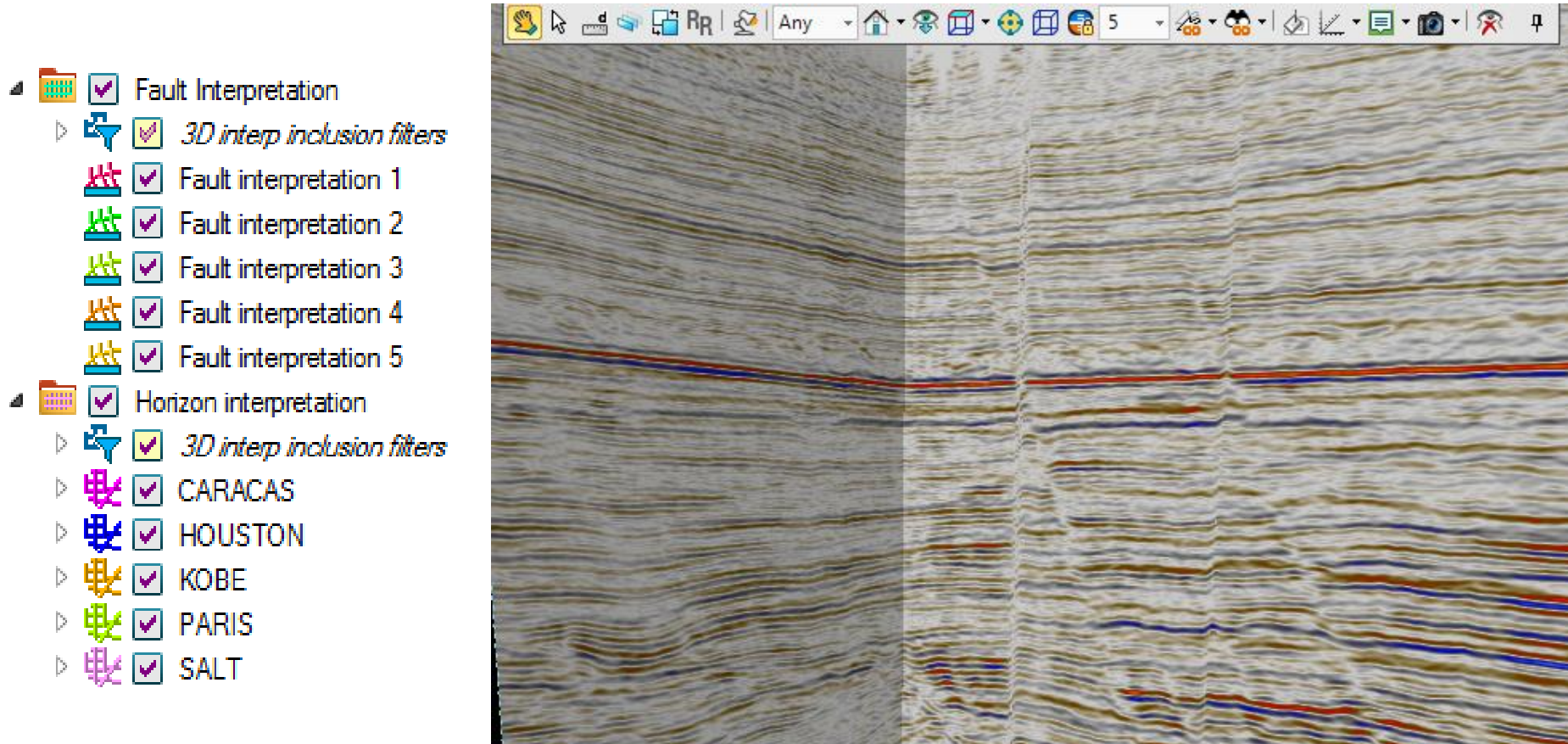
Flatten 3D volumes (1)

1. Reference surface from where the cube was flattened.
2. Level of the flattening.
3. Objects to convert:
 - a. Surfaces
 - b. Horizons
 - c. Faults.
4. Click *Flatten* to shift the objects.



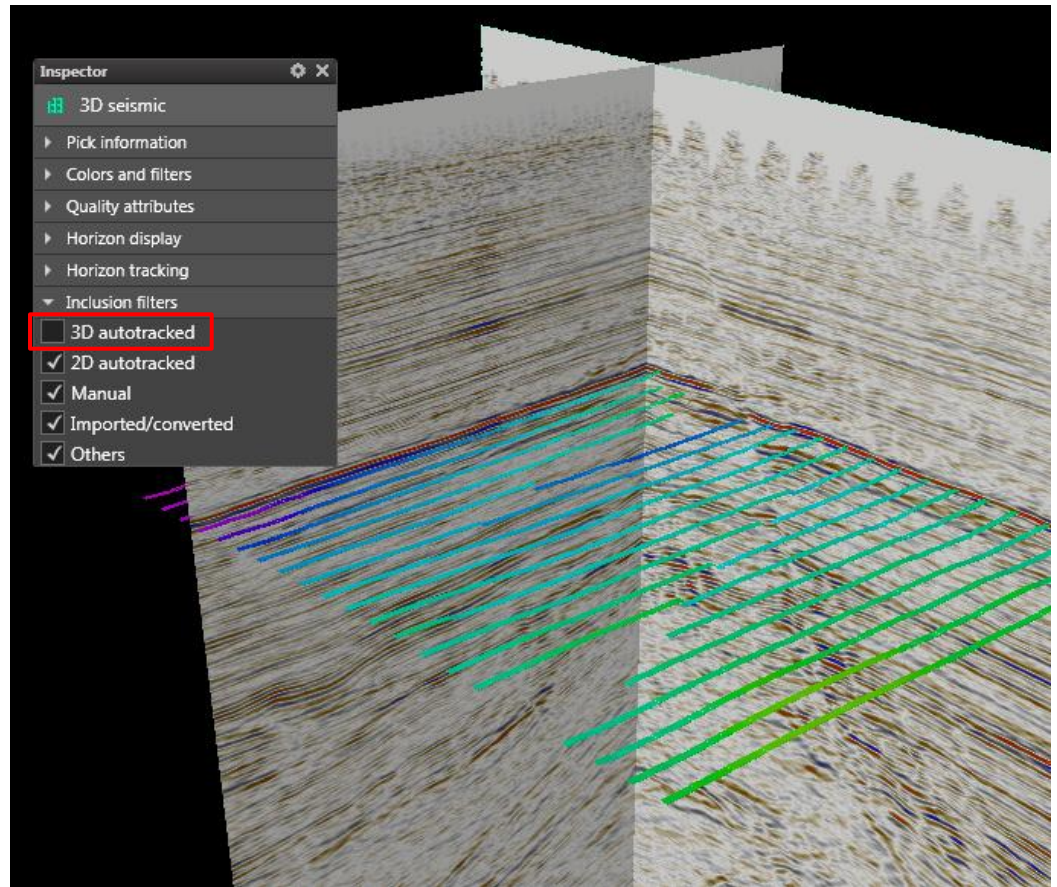
Flatten 3D volumes (2)

Objects in the list can be *Re-flatten*, *Un-flatten*, and *Flatten*.



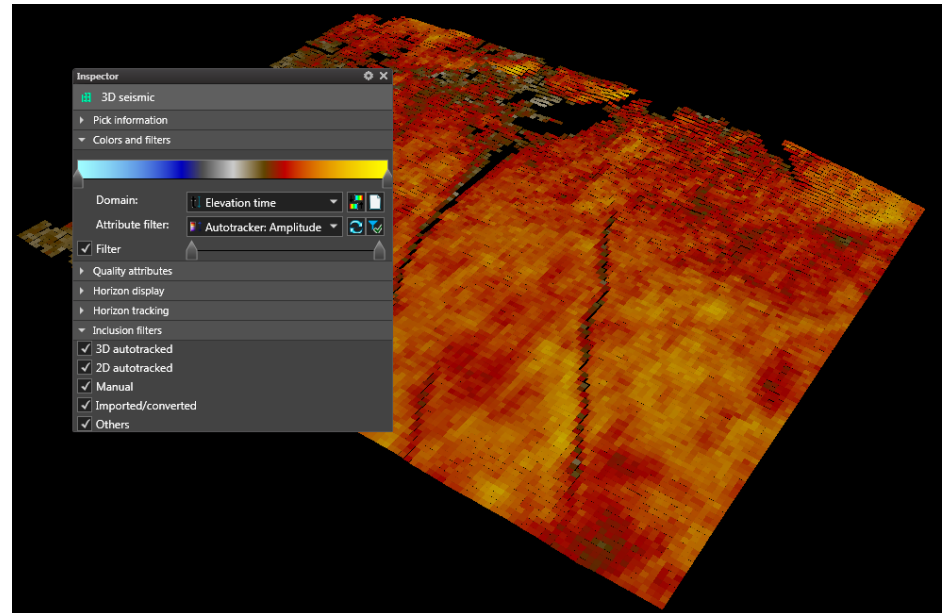
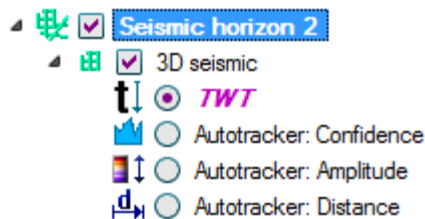
Interpretation filters (1)

- Filter the interpretation on different parameters.
- Use the filters to display the interpretation and to define points as seeds for autotracking and locking points or to select them for reinterpretation.
- Use the **Inspector** to access these filters.



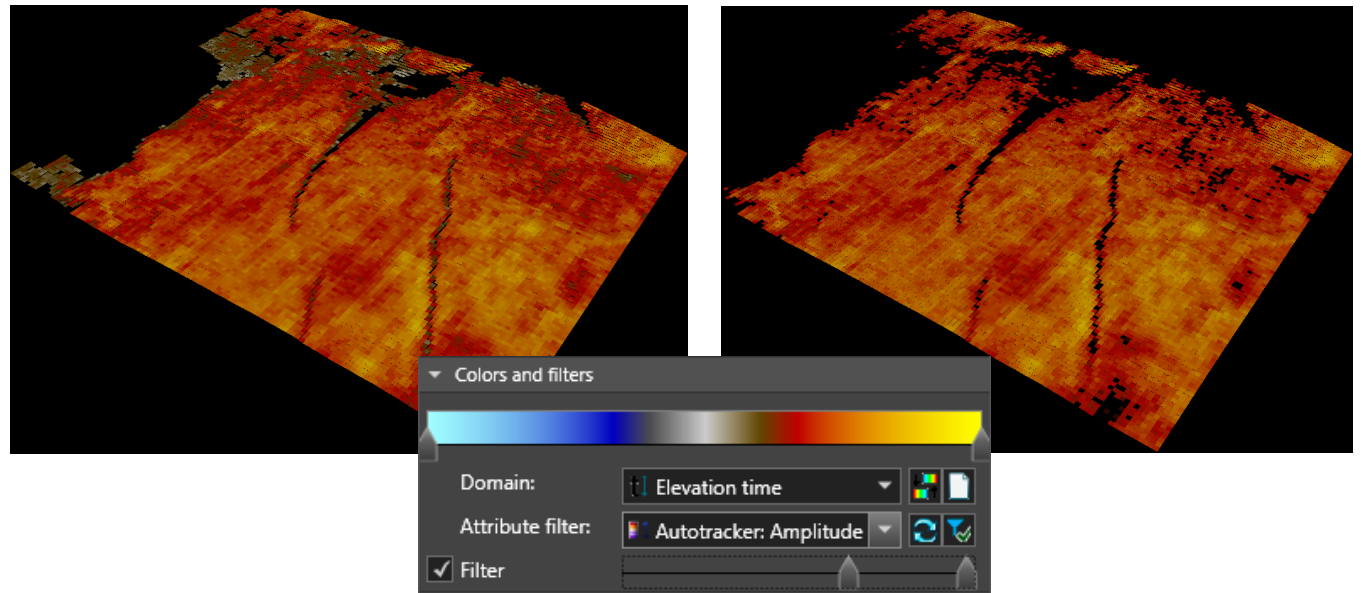
Interpretation quality filters (2)

1. From the **Input** pane, under the horizon interpretation, display the output attributes created during autotracking.
 2. In the **Inspector** under the **Colors and filters** section, select the check box next to *Filter* and choose one attribute to filter in the *Attribute filter* list.
- When you do this, the display updates.



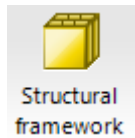
Interpretation quality filters (3)

3. Move the slider next to *Filter* to filter it away from the display.
4. To delete the filtered section click *Run filter* . There is no Undo when this action is performed, so it is best to keep a copy of the original interpretation.

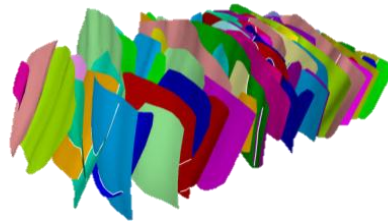


Structural framework workflow: Horizon modeling

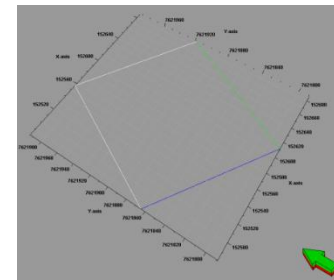
**Define the
Structural
framework**



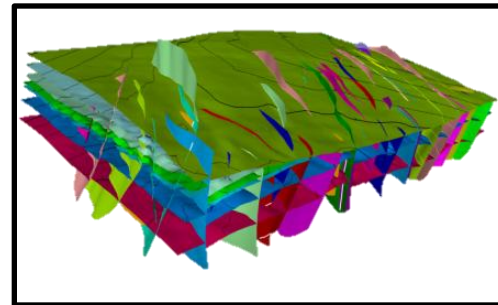
**Fault framework
modeling**



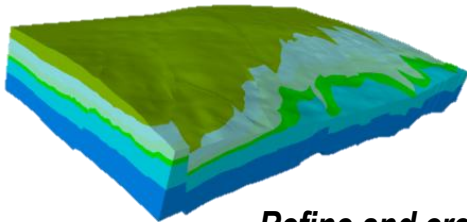
**Boundary
Definition**



**Unrefined Horizon
modeling
(VBM step 1)**



**Refine and create zone
model (VBM step 2)**



Horizon modeling settings: Sequence numbering (1)

- The horizons are grouped into conformable sequences that are determined by horizon type, based on unconformities.

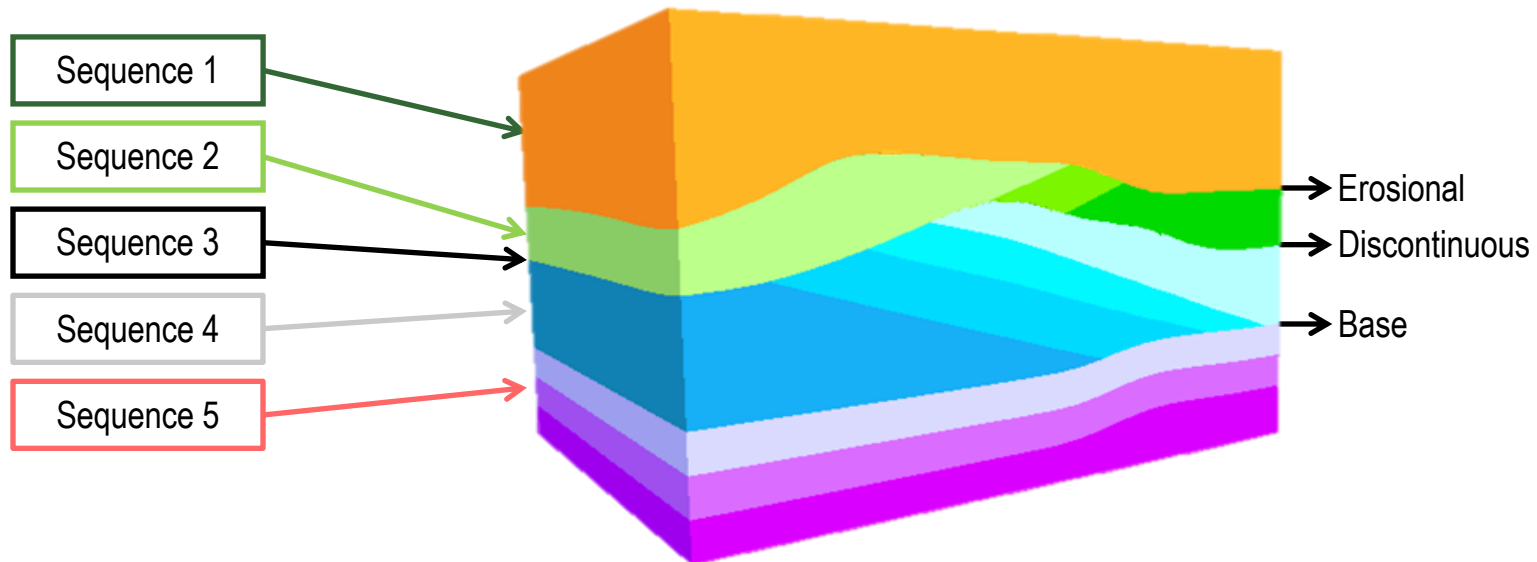
The screenshot displays the 'Horizon modeling' software window. The 'Hints' tab is active, showing 'Edit existing: Structural model' and 'Horizon modeling technique: Volume based modeling'. A warning icon indicates 'Unrefined model : coarse resolution'. The 'Compute horizons' button is highlighted with a red box. Below it, a table lists three horizons: 'Base Cretaceous [Filtered]', 'Upper Jurassic 1 [Filtered]', and 'Upper Jurassic 2 [Filtered]'. The 'Sequence' column shows values 1, 2, and 2 respectively. The 'Horizon type' column shows 'Erosional' and 'Conformable'. The 'Conforms to' column shows 'None' for all. The 'Smoothness' column shows '35' for all. The 'Well tops', 'Non-filtered data', and 'Isochore' columns show icons for each horizon. The 'Input #1' column shows the corresponding horizon names. The 'Sequence' and 'Horizon type' columns are highlighted with red boxes.

Index	Horizon	Sequence	Color	Status	Horizon type	Conforms to	Smoothness	Well tops	Non-filtered data	Isochore	Input #1
1	Base Cretaceous [Filtered]	1		New	Erosional	None	35				Base Cretaceous [Filtered]
2	Upper Jurassic 1 [Filtered]	2		New	Conformable	None	35				Upper Jurassic 1 [Filtered]
3	Upper Jurassic 2 [Filtered]	2		New	Conformable	None	35				Upper Jurassic 2 [Filtered]

Horizon modeling settings: Sequence numbering (2)

Example of a model that contains 5 sequences:

Index	Horizon	Sequence	Color	Status	Horizon type
1	H1	1		✓ Cha...	Erosional
2	H2	2		✓ Cha...	Conformable
3	H3	2		✓ Cha...	Conformable
4	H4	3		✓ Cha...	Discontinuous
5	H5	4		✓ Cha...	Conformable
6	H6	4		✓ Cha...	Conformable
7	H7	4		✓ Cha...	Conformable
8	H8	5		✓ Cha...	Base
9	H9	5		✓ Cha...	Conformable
10	H10	5		✓ Cha...	Conformable

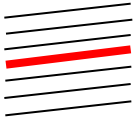


Horizon type and sequence (1)

- The analysis and assignment of the right Horizon type, and the resulting conformable sequences, is fundamental in structural modeling, in particular in the VBM algorithm. Conflicts between input data and horizon type (for example, truncating interpretation for conformable horizons) can result in bad modeling results.
- Horizon types are either Conformable horizons or unconformities (Erosional, Base, and Discontinuous horizons) and reflect the stratigraphic relationship they have with other horizons; that is, the possible termination of other horizons (truncation/topcut or onlap/downlap) against the horizon.
- Unconformities are limiting conformable Sequences, which are a succession of conformable horizons. By definition, horizons within a conformable sequence do not present any truncations (zone pinchout).
- Horizon types and the resulting truncations have an effect on the Stratigraphic layering of Structural grids derived from a Structural framework model.

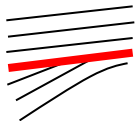
Horizon type and sequence (2)

Conformable



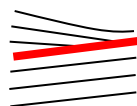
- By definition, a Conformable horizon does not present any stratigraphic terminations against it. However, Conformable horizons themselves can terminate against unconformities.
- By definition, conformable horizons do not intersect. However, they are not necessarily parallel and can present large thickness variations.

Erosional



- By definition, stratigraphically lower horizons can (but do not need to) terminate against an Erosional horizon (truncation/topcut terminations are allowed).
- By definition, an Erosional horizon is conformable to stratigraphically higher horizons (onlap/downlap terminations are prohibited).
- By definition, an Erosional horizon is the base of a conformable sequence.

Base



- By definition, stratigraphically higher horizons can (but do not need to) terminate against a Base horizon (onlap/downlap terminations are allowed).
- By definition, a Base horizon is conformable to stratigraphically lower horizons (truncation/topcut terminations are prohibited).
- By definition, a Base horizon is the top of a conformable sequence.

Discontinuous



- By definition, stratigraphically lower horizons can (but do not need to) terminate against a Discontinuous horizon (truncation/topcut terminations are allowed).
- By definition, stratigraphically higher horizons can (but do not need to) terminate against a Discontinuous horizon (onlap/downlap terminations are allowed).
- By definition, a Discontinuous horizon does not conform to any other horizon and, therefore, represents a conformable sequence on its own.

Seismic Sequence Stratigraphy - Reflection Terminations

- Onlap
- Downlap
- Toplap
- Truncation

